A PCA and K-Means Clustering Analysis of Offensive Play-Caller Archetypes

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Abstract

This project develops a quantitative framework for classifying NFL offensive coordinators by reducing modern offensive design into three interpretable dimensions derived from twelve formation, personnel, and situational play-calling features. Using league-wide play-by-play and alignment data from 2022–2025, principal component analysis (PCA) yielded three components capturing (1) passing aggression and early-down pass tendency, (2) spread versus heavy-personnel alignment, and (3) tempo-driven aggressiveness. K-means clustering revealed three distinct archetypes: spread-run systems, balanced modern passing attacks, and under-center heavy-personnel offenses. This multidimensional approach provides a scalable method for profiling offensive coordinators, evaluating year-to-year consistency, and understanding how schematic archetypes relate to performance and advancement.

Introduction

Offensive football in the modern NFL features unprecedented structural diversity and philosophical variation. Teams differ not only in pass-run balance, but in deeper architectural decisions including formation width, personnel grouping, pre-snap structure, situational aggression, and tempo. While analytics has advanced play outcome evaluation through metrics like EPA and success rate, far less work has quantified the underlying schematic tendencies differentiating offensive coordinators. Discussions of offensive identity often rely on broad qualitative labels including West Coast, spread and pro style that lack formal measurement.

This project develops a scalable, data-driven framework for classifying offensive coordinators based on structural and philosophical tendencies. The goal is measuring the underlying choices defining offensive design, not evaluating performance. By integrating formation usage, personnel deployment, and situational play-calling into a unified dataset, we construct a multidimensional representation of offensive identity grounded in observable pre-snap decisions.

Principal component analysis extracts the dominant axes of variation, yielding three components capturing passing aggression, spread versus heavy alignment, and tempo. Each coordinator-season becomes a point in three-dimensional identity space, enabling direct comparison across coordinators, teams, and years. K-means clustering then identifies natural groupings revealing stable, interpretable offensive styles corresponding to recognizable football systems. This framework provides a rigorous foundation for understanding offensive design,

enabling objective scheme comparisons and offering a new perspective on how offensive identity aligns with team success and coordinator advancement.

Literature Review

Quantitative analysis of offensive football strategy has expanded as detailed tracking and alignment data became accessible. Traditional football literature categorizes offensive schemes qualitatively (west coast, spread, option)emphasizing how formations, personnel, and play structures create space and leverage matchups. These frameworks focus on structural elements including formation width, blocking concepts, and pre-snap alignment.

Within analytics, more formal attempts have quantified offensive identity. Pro Football Focus conducted league-wide studies examining how offensive tendencies correlate with efficiency, identifying differences in passing aggression, run frequency, and situational behavior. However, this work often blends pre-snap identity with post-snap outcomes, obscuring the philosophy driving coordinator decisions.

Independent analytics research has used clustering to group NFL offenses based on pass rate over expected, explosive play frequency, and situational success rates. M-Football Analytics applied unsupervised learning to generate offensive style clusters using outcome-driven and rate-based variables, demonstrating clustering's potential. However, these studies primarily cluster whole teams and rely heavily on efficiency results rather than isolating architectural choices defining offensive structure.

A key distinction in this study is the exclusive use of pre-snap variables rather than outcome-based measures. Every variable (formation width, personnel grouping, shotgun versus under-center alignment, down-and-distance tendencies, tempo) is measured before the snap. This separates offensive identity from performance, allowing clustering to characterize how coordinators construct and sequence their offense independent of play success. This addresses a limitation where outcome-based features conflate scheme with execution quality, quarterback play, or defensive matchup.

By reducing offensive structure into interpretable pre-snap variables and applying dimension reduction at the coordinator level, this study extends previous work, contributing a more precise, scheme-focused model of offensive identity comparable across coordinators, teams, and seasons without performance-driven noise.

Data and Feature Engineering

Data Sources

This study integrates three complementary datasets to construct comprehensive pre-snap variables describing offensive identity. All variables reflect decisions made before the snap, isolating coordinator philosophy rather than execution quality.

Formation Data (Sumer Sports):

Formation types defined using two-number notation (2x1, 2x2, 3x1, 1x3), where numbers indicate eligible receivers aligned left and right of the offensive line. Dataset includes usage rate, pass rate, and deviation from league-average formation usage, contributing to understanding of coordinator use of width, spacing, and alignment.

Personnel Data (Sumer Sports):

This tracks personnel grouping usage describing distribution of five non-linemen, non-quarterback skill players across running backs, tight ends, and wide receivers. Notation uses two digits, the first representing running backs (including fullbacks), the second representing tight ends. Wide receivers are calculated by subtracting from five. Dataset includes usage rate, pass rate, and league deviation, capturing how coordinators structure offenses via skill-position configurations.

Play-by-Play Data (NFLfastR):

Detailed records covering every offensive snap. Only pre-snap variables used: down, play type, run direction, pass depth, quarterback alignment (shotgun vs. under center), and tempo (huddle vs. no-huddle). The variables are aggregated into season-level indicators including down-and-distance pass rates, fourth-and-short attempt frequency (aggression measure), shotgun usage percentage, and no-huddle rate (tempo indicator).

After collection, relevant pre-snap variables were merged and reduced to 12 distinct structurall features representing formation width, personnel density, situational pass tendencies, tempo, and alignment choices, serving as input to unified PCA.

Feature Construction

Originally, the feature set included every personnel, formation, and situational variable. Initial PCA on this full space failed to capture meaningful variance proportion, indicating substantial redundancy. The set was first reduced to 18 variables, but PCA still showed limited variance explained. After further refinement and correlation analysis, the set was narrowed to 12 core features, removing redundancy while retaining meaningful offensive behavior components.

During this process, personnel usage explained significantly more about offensive identity than formation variables, which proved noisy and less predictive. All formation-based features were dropped for purely personnel-driven representation.

Personnel groupings collapsed into three standardized buckets based on wide receivers on the field:

- 0–1 WR (heavy personnel)
- 2 WR (balanced personnel)
- 3–5 WR (spread personnel)

This preserved strategic meaning while reducing dimensionality. For year-to-year comparability, only two personnel variable types were retained: league-difference measures capturing OC deviation from league norms, and weighted pass-rate metrics for each WR group summarizing aggressive usage in passing.

For situational playcalling, only pass-related variables were kept, as run variables are inverse of pass tendencies and therefore redundant. Overall pass percentage was excluded as it provides no information beyond granular situational breakdowns. The following targeted pass-rate variables were used:

- Pass rate on 1st down
- Pass rate on 3rd down
- Pass rate from shotgun
- Pass rate under center
- Percent of plays from shotgun vs. under center
- Percent of plays in no-huddle (tempo proxy)

This final set is compact, non-redundant, strategically interpretable, and stable across seasons. These features are well-suited for clustering coordinators and evaluating playcalling identity changes over time.

PCA Construction

Following feature consolidation, 12 pre-snap variables were standardized and subjected to a single principal component analysis. The cumulative scree plot (Figure 1) shows that the first three principal components explain approximately 57.6% of total variance. Although 57.6% may seem modest, this is notably strong for NFL analytics. Offensive structure is highly complex and contextually influenced, so achieving over half the variance with only three dimensions indicates clear, coherent underlying patterns in coordinator behavior. These three components represent dominant axes of offensive coordinator identity variation and serve as foundation for clustering analysis.

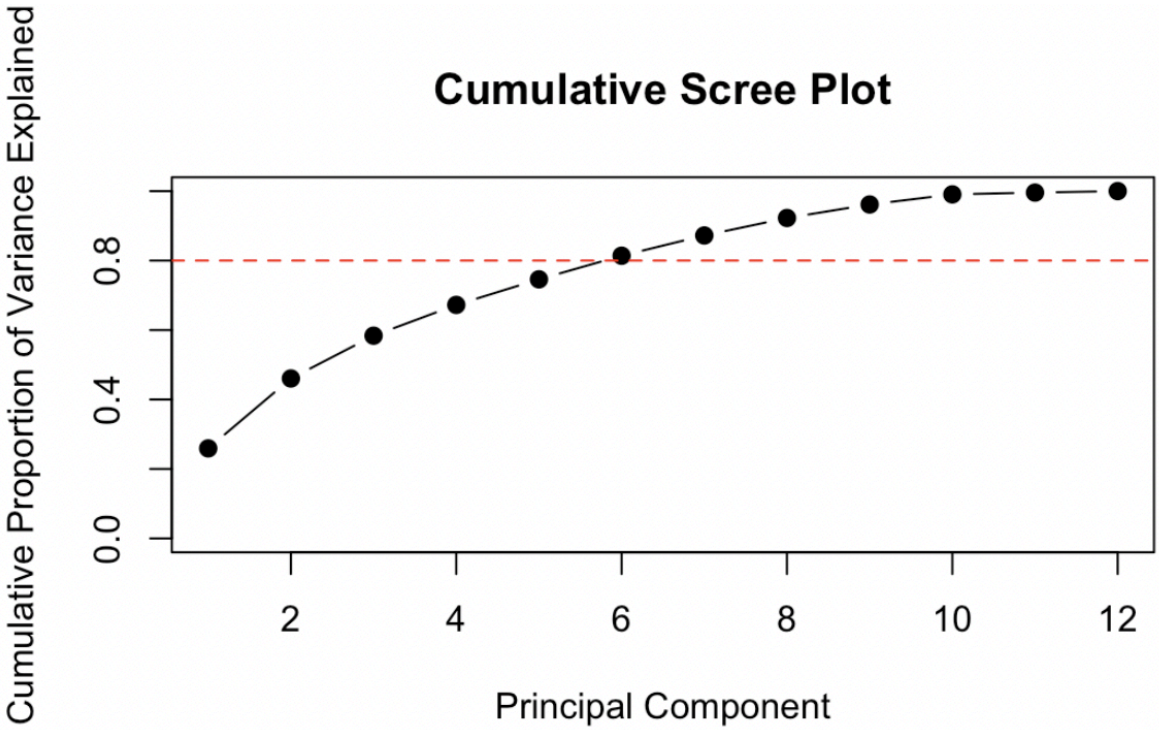


Figure 1: Cumulative Scree Plot

Results

PCA Results

PC1: Overall Passing Identity

PC1 loads positively on variables associated with passing volume and aggression: pass rate from shotgun, pass rate on first down, pass rate on third down, and pass rate under center. It loads negatively on league-adjusted usage of heavier personnel groupings (0–1 WR and 2 WR structures).

High PC1: Pass-heavy coordinators, aggressive early-down passing, heavy shotgun usage, less reliance on compressed formations or heavy personnel.

Low PC1: Run-leaning or balanced coordinators, traditional sequencing on early downs, greater use of heavier, compressed personnel.

PC1 represents overall passing identity: a broad axis distinguishing pass-first coordinators from run-oriented or balanced systems, forming the primary philosophical dimension of offensive design.

Figure 2 shows the top 3 and bottom 3 values for PC1. One thing that is very interesting is that the Atlanta Falcons completely changed their scheme for PC1 from 2023 to 2024. This is immediately after they switched offensive coordinators from Dave Ragone to Zac Robinson.

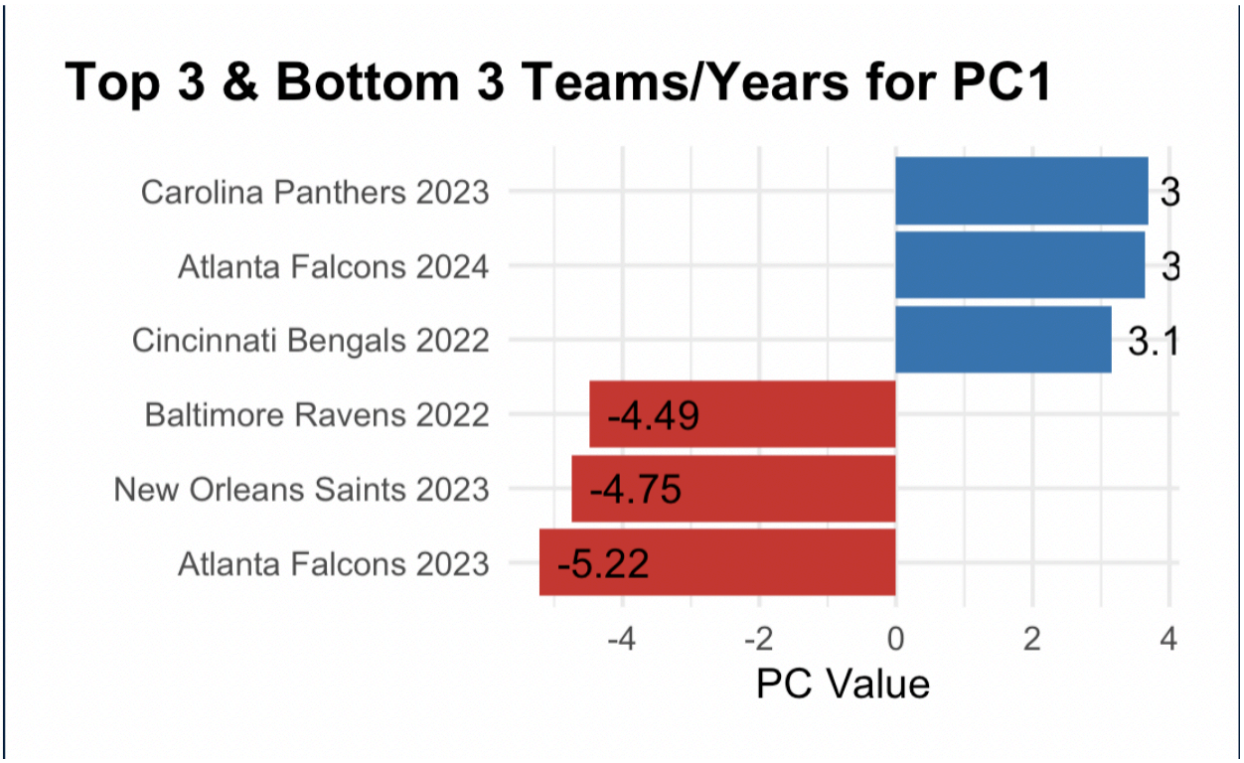


Figure 2: Top 3 and Bottom 3 Teams for PC1

PC2: Spread vs. Heavy Personnel Identity

PC2 was sign-flipped for interpretability. After flipping, positive loadings correspond to spread-oriented structure, negative to heavy-personnel structure. PC2 loads positively on shotgun usage and 3-4 WR spacing indicators, negatively on under-center usage and differences favoring 1-WR or 2-WR formations.

High PC2: Wide formations with 3-4 WRs, shotgun-based alignment, spread spacing and distribution, reduced use of under-center or heavy structures.

Low PC2: Frequent 12 and 21 personnel, under-center alignment, tight condensed formations, traditional pro-style grouping patterns.

PC2 captures formation width and spacing, functioning as a spread-versus-heavy personnel spectrum.

Figure 3 shows the top 3 and bottom 3 teams for PC2. One thing that is very interesting is the Baltimore Ravens were in the bottom 3 for the years 2022 and 2024. They had two different coordinators during this time period. This shows that an offensive identity isn't just based off the offensive coordinator.

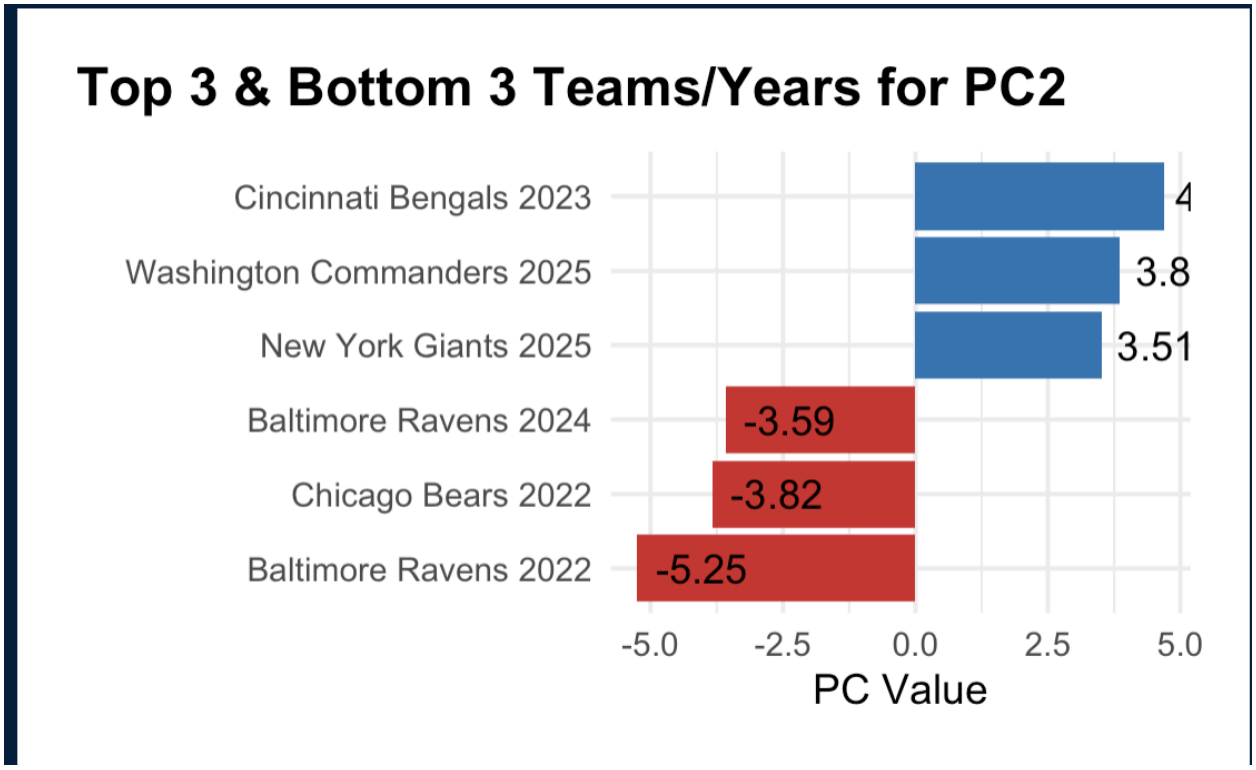


Figure 3: Top 3 and Bottom 3 Teams for PC2

PC3: Tempo and Early-Down Aggressiveness

PC3 loads strongly on shotgun usage, no-huddle rate, and pass rate on first down. It loads negatively on pass rate from compressed 0-1 WR structures.

High PC3: Faster tempo and increased no-huddle, shotgun pace and rhythm-based sequencing, aggressive early-down passing, greater reliance on spread passing concepts.

Low PC3: Slower, more deliberate pace, traditional sequencing with less no-huddle, lower early-down pass aggression, heavier, more compressed passing structures.

PC3 represents tempo and situational aggressiveness, distinguishing coordinators using pace and early-down initiative from those relying on slower, traditional sequencing.

These three components define a three-dimensional offensive identity space where every OC-season can be precisely located.

Figure 4 shows the top 3 and bottom 3 teams for PC3. The high values are all teams in 2024 and 2025 and the low values are all teams in 2023 and 2024. This signals a potential shift towards faster and more aggressive offenses.

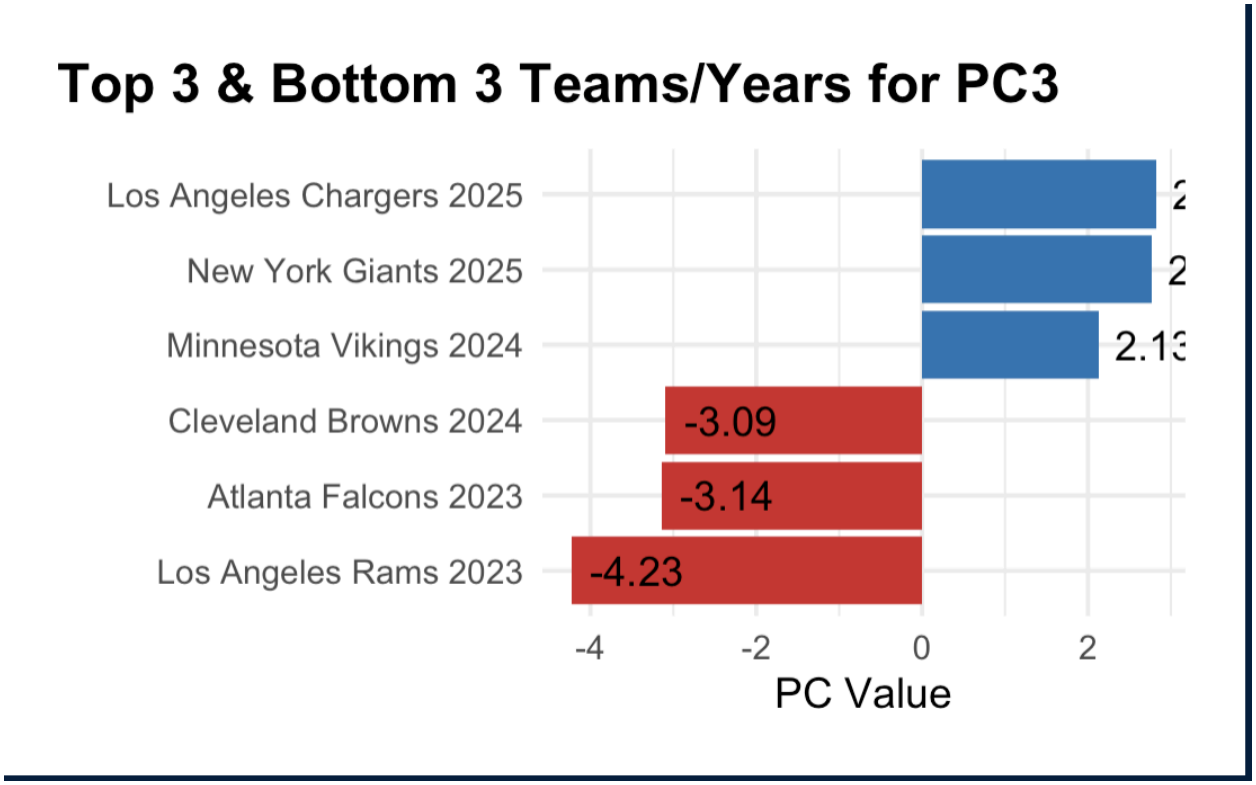


Figure 4: Top 3 and Bottom 3 Teams for PC3

Summary of Principal Components

Principal Component	Name	High Values	Low Values
PC1	Overall Passing Identity	Pass-heavy, early-down aggression, shotgun passing	Run-heavy, balanced, compressed formations
PC2	Spread vs Heavy Personnel	Spread spacing, 3-4 WR, shotgun allignment	Heavy personnel, under-center, tight formations

PC3	Tempo and Aggressiveness	No-huddle, fast pace, shotgun tempo, early down passing	Slow tempo, traditional sequencing, heavy play structures

Table 1: Summary of Principal Components

Cluster Analysis

After projecting each OC-season into three-dimensional PCA space, k-means clustering identified natural groupings of offensive coordinator philosophies. The elbow plot (Figure 5) shows clear inflection at k=3, indicating a three-cluster solution captures major structural and philosophical distinctions while maintaining parsimony and interpretability.

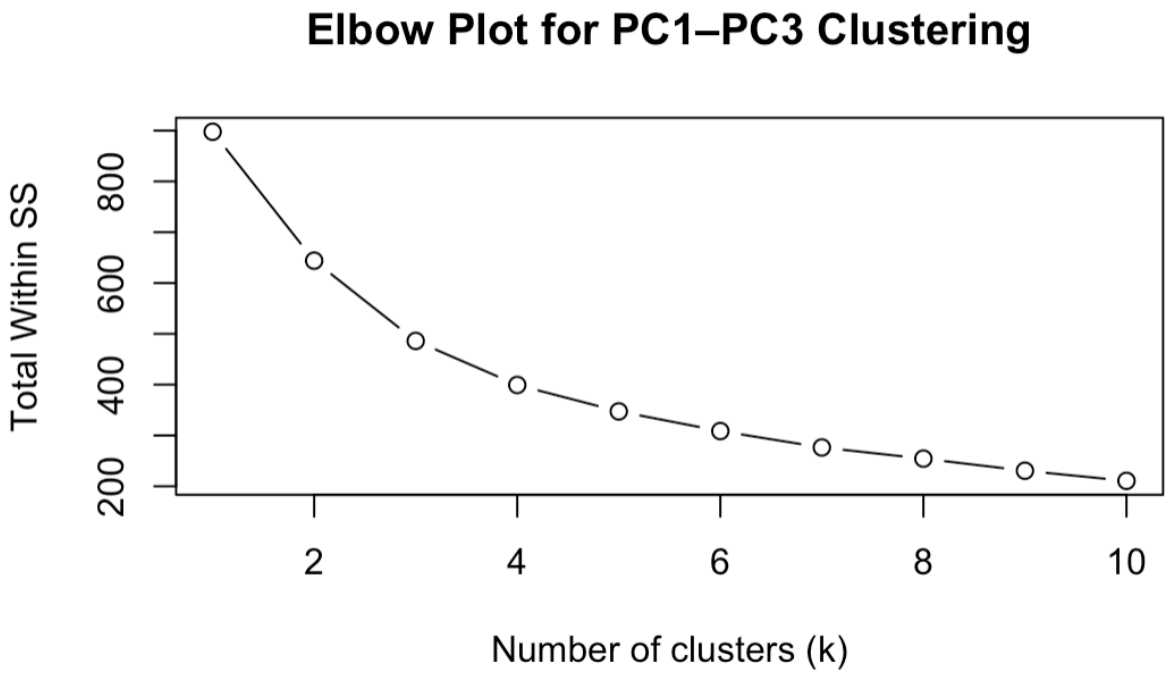


Figure 5: Elbow plot for clustering

The resulting clusters represent three coherent offensive archetypes grounded in pass-identity (PC1), spread-versus-heavy offense (PC2), and tempo-aggressiveness (PC3) axes, demonstrating distinct schematic families suggesting coordinators exhibit meaningful, persistent structural tendencies across seasons.

The interactive plot shows that there are three distinct well separated clusters in the 3 dimensional space. The 2022 Ravens are in cluster 3 due to their low values for PC1 and PC2.

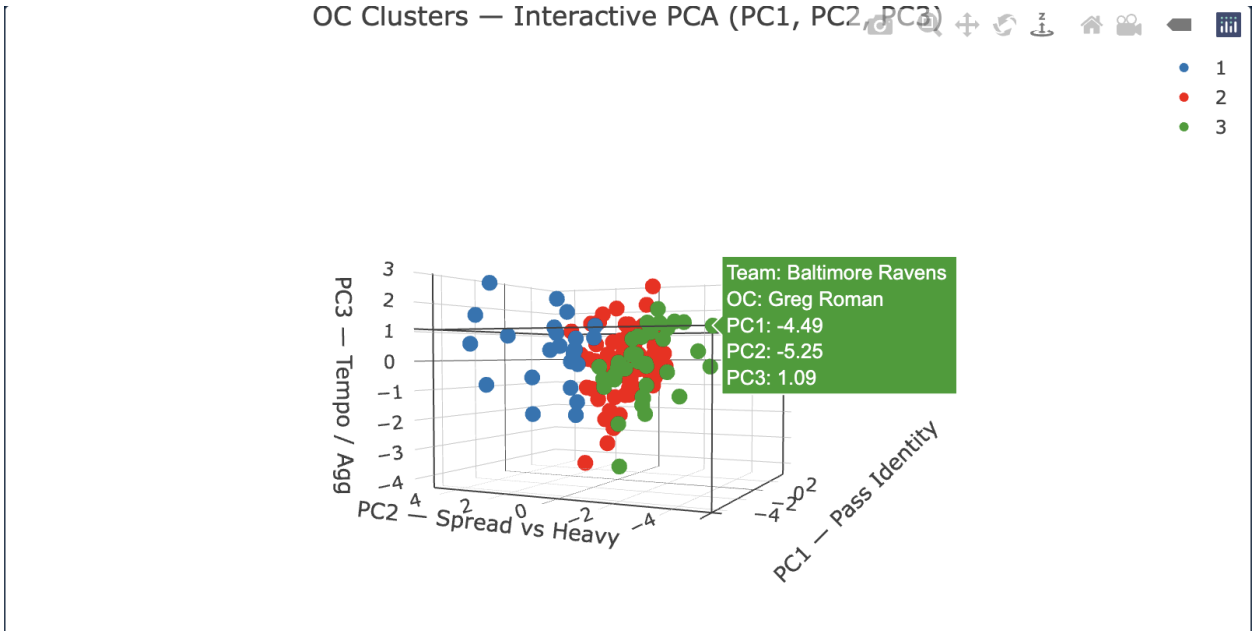


Figure 6: 3D Cluster

Cluster 1: Spread-to-Run (PC1: –2.11, PC2: +1.73, PC3: +0.39)

This cluster represents offenses that look spread but behave run-first. These coordinators employ wide spacing, shotgun formations, and 3–4 WR sets, yet lean heavily into running. Spread geometry forces lighter boxes and horizontal stress, creating structural advantages for the run game. Moderate tempo supports rhythm-based sequencing without becoming a full tempo offense.

Key traits: Heavy use of spread formations and horizontal width, shotgun-centered alignment, run-first sequencing despite spread structure, moderate no-huddle/tempo elements.

Examples: Arthur Smith (Titans/Falcons), Kyle Shanahan wide-zone variants, Mike McDaniel in select scripts.

Cluster 2: Modern Balanced Spread (PC1: +1.23, PC2: +0.29, PC3: –0.10)

Comprises standard modern passing coordinators. These offenses lean pass-first, operate primarily from shotgun, and incorporate spread elements, but don't rely heavily on extreme tempo or personnel specialization. They represent the league's dominant passing archetype, emphasizing early-down aggression and efficient spacing without extreme structural tendencies.

Key traits: Pass-first philosophy, shotgun base with moderate spread alignment, balanced structure within pass-heavy framework, neutral pace and controlled sequencing.

Examples: Kellen Moore, Ben Johnson, Shane Steichen, Kevin O'Connell.

Cluster 3: Traditional Pro-Style (PC1: –0.94, PC2: –1.61, PC3: –0.07)

Represents traditional pro-style offensive identity. These coordinators rely heavily on 12 and 21 personnel, under-center snaps, and condensed formations. Sequencing tends to be slower and more deliberate, with run-first orientation and less early-down passing. Structure emphasizes physicality, play-action, and formation versatility grounded in heavier personnel usage.

Key traits: Frequent 12/21 personnel and tight formations, high under-center usage, slower tempo and conservative sequencing, run-first orientation with reduced passing variability.

Examples: Matt LaFleur (under-center variants), early-career Sean McVay, Greg Roman (Ravens/49ers), Arthur Smith (late Titans era).

The three-cluster solution identifies three dominant strategic archetypes in modern NFL offenses: spread-to-run systems leveraging width and spacing to support run-first schemes, modern spread passing offenses representing contemporary pass-first models, and traditional heavy-personnel pro-style offenses built around under-center structure and run-game physicality. Despite NFL play design complexity, offensive coordinators consistently operate along a limited set of structural dimensions shaped by formation width, passing aggression, and tempo philosophy.

Summary of Clusters

Cluster	Style Name	Key Traits
1	Spread-to-Run Offense	Spread Formations, run-first identity, shotgun base, moderate tempo
2	Modern Balanced Spread	Pass-heavy, balanced spread, standard modern shotgun offense
3	Traditional Pro-Style	Pro-style, heavy sets, under-center runs, slower pace

Table 2: Summary of Clusters

Application of Clusters

The three PCA-derived clusters represent distinct coordinator archetypes whose real-world implications can be evaluated through performance metrics, coaching outcomes, and year-to-year scheme stability.

Cluster Differences in Offensive Performance

The boxplot of points per possession (Figure 7) reveals meaningful performance differences. Cluster 1 coordinators score the fewest points per possession, with noticeably lower median compared to other groups. Cluster 2 offenses demonstrate moderate efficiency, generally outperforming Cluster 1 while showing slightly higher variability. Cluster 3 offenses exhibit the most stability. They have a similar median to cluster 2 but lower variation. This suggests stylistic tendencies captured by PCA relate to measurable on-field output. More aggressive passing structures and modern personnel usage appear associated with better scoring efficiency.

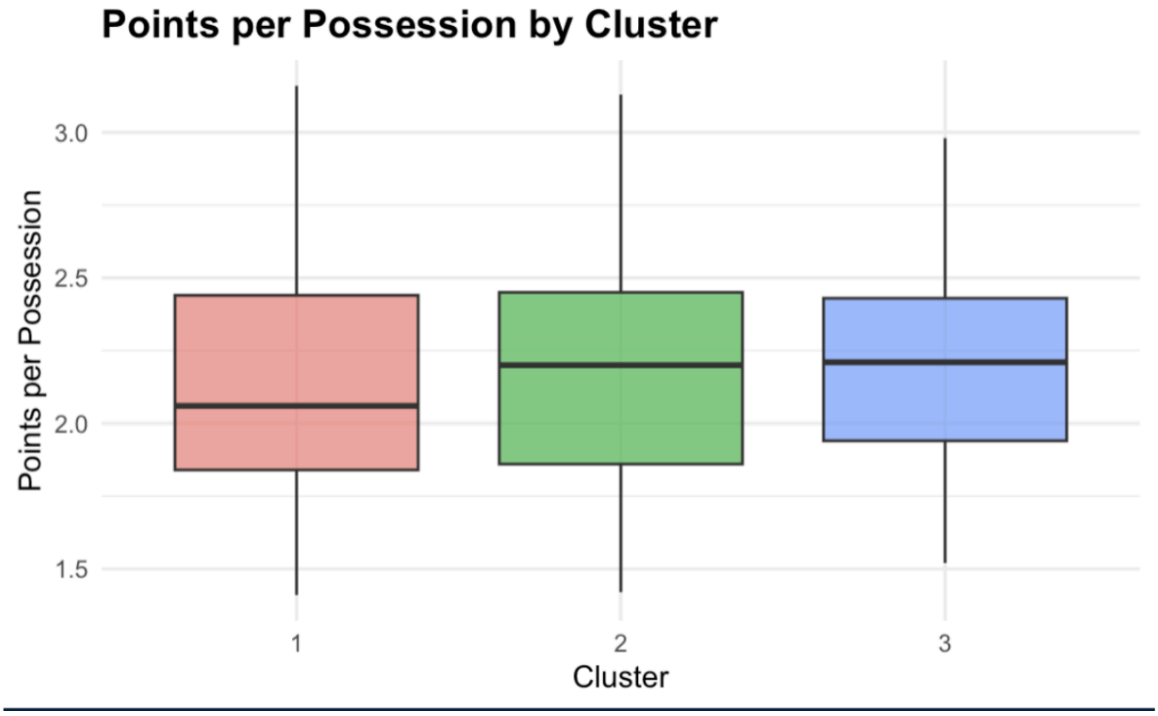


Figure 7 : Boxplot of points per possession by cluster

Career Outcomes by Cluster

Career trajectory patterns (Figure 8) provide further validation. Cluster 1, corresponding to lowest scoring efficiency, shows the highest proportion of demotions or negative outcomes. Cluster 2 presents the most balanced distribution, with moderate proportion of both promotions and stable outcomes. Cluster 3, despite scoring highest, exhibits a very large share of "same

role" outcomes, indicating high-performing coordinators often maintain positions rather than jumping immediately to new opportunities. This is not a bad thing. For such a volatile position, stability is a great measure of success. Interestingly, promotions occur most frequently in Cluster 2, suggesting these coordinators combine solid performance with enough perceived growth potential to attract higher-level roles.

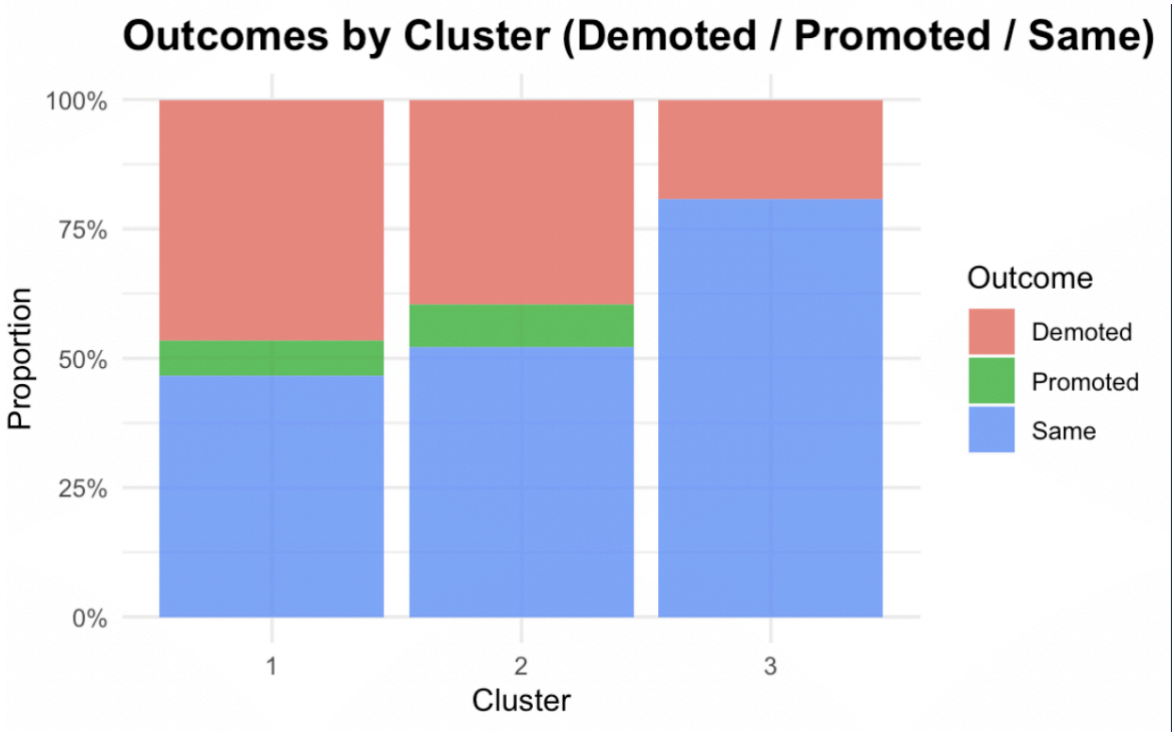


Figure 8: Career Outcomes by Cluster

Distribution of Clusters by Season

Year-by-year breakdowns show clusters are not static but shift as offensive trends evolve league-wide. This is shown in Figure 9. Cluster 2 is consistently the largest group across all four seasons, indicating balanced offensive structures remain the most common NFL approach. Cluster 3 grows slightly over time, reflecting the league's gradual movement toward spread personnel, shotgun-heavy usage, and more pass-driven schemes. Cluster 1 fluctuates but remains the smallest group, consistent with declining prevalence of heavy-personnel, conservative offenses. This trend aligns with known league-wide movement toward efficiency-driven, pass-first approaches.

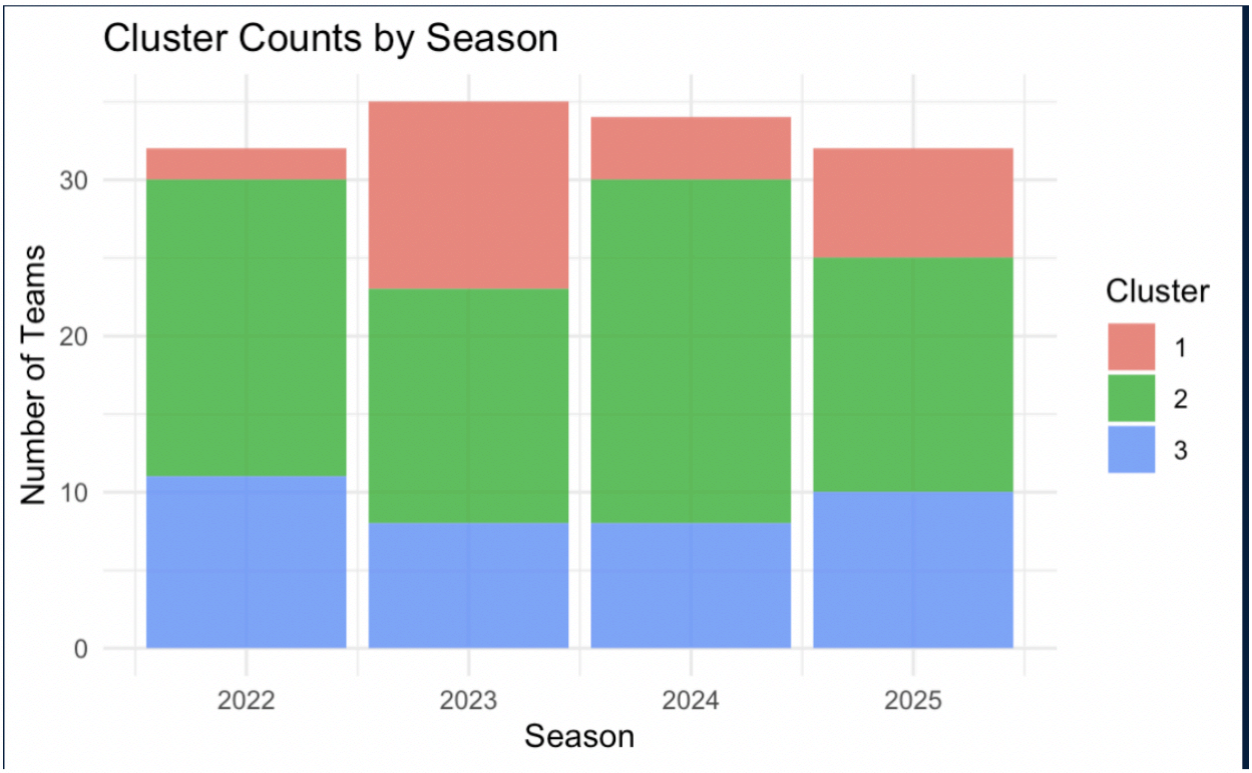


Figure 9: Cluster Counts per Season

Probability of Scheme Change

Scheme continuity is critical for organizational planning, and cluster outputs provide insight into how often systems are reset year to year (Figure 10). Teams whose offensive coordinator was fired had roughly 40% chance of a new scheme the following season. Teams whose coordinator was retained had a similar but slightly higher rate (42.6%). League-wide, overall baseline probability of scheme change was 41.7%. This indicates scheme turnover is not driven solely by offensive coordinator changes. Many offenses switch systems adapting to personnel changes, roster composition, and evolving league trends, even when coordinator remains employed. The clusters therefore capture not just who is calling plays, but how strategic identity shifts across seasons.

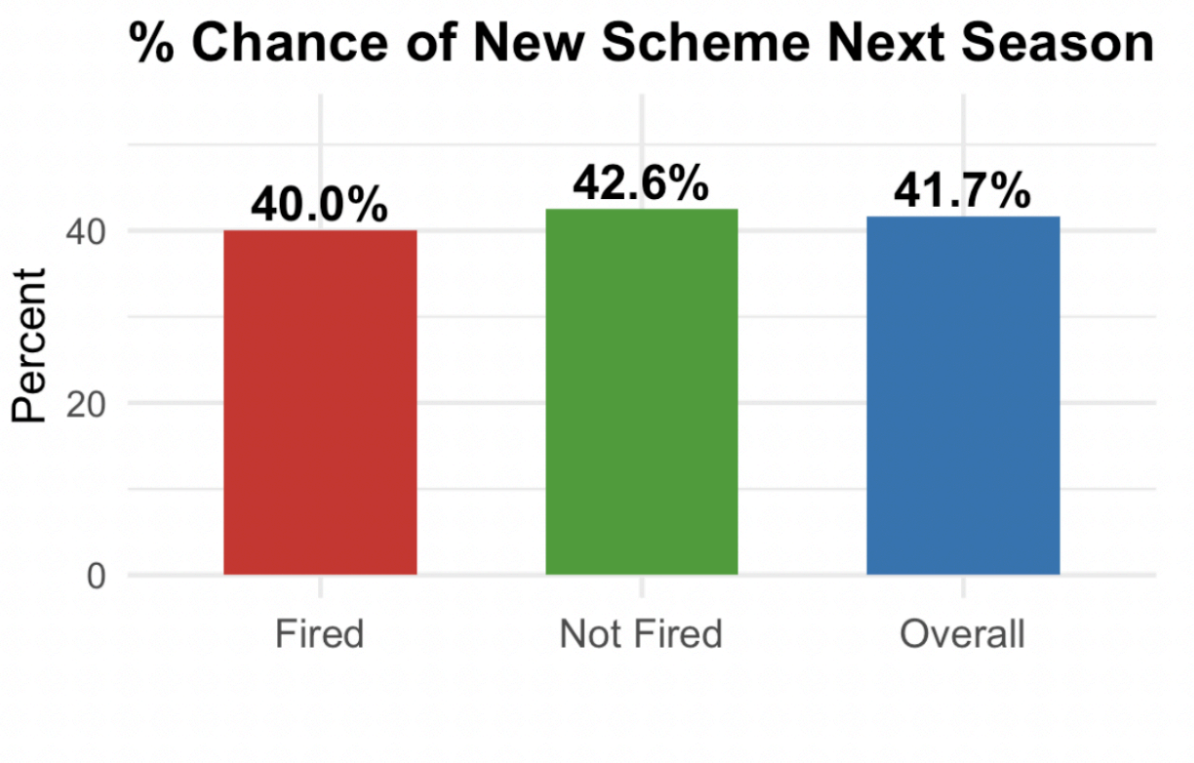


Figure 10: Percent chance of a new scheme the next season

Discussion

The results demonstrate that structural tendencies of NFL offensive coordinators can be meaningfully reduced to three core dimensions: overall passing identity, spread versus heavy personnel usage, and tempo-based aggressiveness. Despite inherent complexity of offensive design, PCA captured over half of total variance with just three components, revealing coordinator strategies that follow consistent, quantifiable patterns. These components form a coherent identity space where coordinators cluster into three distinct schematic archetypes.

The clustering analysis highlights that offensive systems naturally group into spread-to-run approaches, modern balanced spread passing offenses, and traditional under-center heavy-personnel structures. Each archetype corresponds to recognizable football concepts and reflects stable differences in how coordinators construct formations, choose personnel, and sequence plays. Clear separation between clusters in the PCA space demonstrates offensive identity is not random or overly individualized but adheres to a small set of structural templates persisting across teams and seasons.

These clusters provide practical value when applied to real-world outcomes. Performance measures such as points per possession, coaching career trajectories, and year-to-year scheme stability all vary systematically across the three groups. More aggressive, spacing-oriented offenses tend to score more efficiently, certain archetypes show higher

promotion or demotion rates, and scheme changes occur frequently even without coordinator turnover. Together, these applications illustrate the PCA-cluster framework not only captures schematic differences but reflects meaningful organizational and on-field consequences.

Conclusion

This project introduced a quantitative framework for identifying offensive coordinator archetypes in the NFL using a streamlined set of personnel and situational passing features. By emphasizing feature reduction and interpretability, the analysis produced stable PCA-based representations capturing meaningful differences in offensive philosophy across seasons.

The resulting clusters provide practical ways to compare offensive styles and track how team identities evolve over time. Rather than relying solely on raw performance metrics, the framework emphasizes strategic tendencies, particularly personnel usage and situational decision making, that shape offensive structure and continuity.

More broadly, this approach demonstrates the value of combining domain knowledge with dimensionality reduction techniques to study coaching behavior. The clustering framework can serve as foundation for future applications, including evaluating scheme fit, anticipating offensive transitions, and extending stylistic analysis to other levels of football.

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Appendix

Dataset 1: Simplified Feature Matrix

	pct_shotgun	pct_no_huddle	pct_pass_shotgun	pct_pass_under_center	pct_pass_first	pct_3rd_pass	0_1_WR_league_diff	0_1_WR_pass_rate	2_WR_league_diff	2_WR_pass_rate	3_4_WR_league_diff	3_4_WR_pass_rate
1	0.75566231983527 8	0.275909402882635	0.66407766990291 3	0.258928571428571	0.48936170212766	0.74891774891774 9	-4.37	34.62	1.36	52.65	6.72	69.997523533671 3
2	0.60692307692307 7	0.130769230769231	0.52133333333333 3	0.236	0.32207207207207 2	0.65803108808290 2	5.27	36.59	-6.31	48.86	1.59	67.141762182041 1
3	0.73646723646723 6	0.039886039886043	0.52258726899384	0.360655737704918	0.43558282208589	0.61061946902654 9	9.49	37.41	0.32	45.57	-11.69	76.05
4	0.62517006802721 1	0.088435374149659 9	0.68989547038327 5	0.351724137931034	0.55681818181818 2	0.71428571428571 4	1.17	39.99	-9.37	65	-4.2	78.751809299801
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16	0.656826 5682656 83	0.0067650 67650676 53	0.709900 9900990 1	0.2840466 92607004	0.535714 2857142 86	0.777777 7777777 78	15.99	19.55	-18.28	46.27	2.96	71.7465 3371776 39
17	0.436915 8878504 67	0.0950155 76323987 4	0.797323 1357552 58	0.3685344 82758621	0.492890 9952606 63	0.735849 0566037 74	-2.35	23.81	-4.45	40.8	3.75	61.7004 0556900 73
18	0.541450 7772020 73	0.0608808 29015544 3	0.791929 3820933 16	0.4077669 90291262	0.536489 1518737 67	0.790441 1764705 88	-5.7	33.33	-17.15	33.95	24.19	65.8399 4536763 03
19	0.436900 3690036 9	0.0369003 69003689 8	0.840867 9927667 27	0.3212121 21212121	0.476293 1034482 76	0.808612 4401913 88	-4.84	21.38	-11.9	49.71	11.35	70.8028 6952987 4
20	0.541310 5413105 41	0.0413105 41310541 3	0.781382 2284908 32	0.3258426 96629213	0.507216 4948453 61	0.875	1.09	20.75	-6.07	42.28	6.490000 00000001	64.7881 2007277 14
21	0.442345 2768729 64	0.1302931 5960912	0.853503 1847133 76	0.4132841 32841328	0.535849 0566037 74	0.860169 4915254 24	8.75	28.11	14.64	47.71	-21.75	63.28
22	0.541571 3196033 56	0.0518688 02440884 8	0.745454 5454545 45	0.2791411 04294479	0.486111 1111111 1	0.738095 2380952 38	1.5	21.18	10.55	47.32	-10.8	68.4419 8978626 82
23	0.540458 0152671 76	0.0503816 79389313 1	0.698940 9984871 41	0.2558139 53488372	0.458904 1095890 41	0.755760 3686635 95	-1.93	19.56	-8.75	42.86	10.5	72.9717 3793859 65
24	0.541479 0996784 57	0.1311897 10610932	0.662835 2490421 46	0.2926829 26829268	0.435114 5038167 94	0.751072 9613733 9	-4.51	47.06	-1.11	46.11	1.13	66.1871 3201320 13

25	0.560551124002901	0.068890500362581 7	0.770401106500692	0.329411764705882	0.497767857142857	0.803347280334728	-4.09	7.42	-18.68	32.84	21.52	72.93969469290 4
26	0.752969121140142	0.161520190023753	0.551174496644295	0.114754098360656	0.458333333333333	0.573033707865169	-4.05	13.03	-14.32	48.1	16.01	72.21654116465 86
27	0.567375886524823	0.096453900709219 4	0.733243967828418	0.177464788732394	0.466386554621849	0.775423728813559	-3.85	46.16	0.41	62.39	-0.060000000000023	73.14721794871 8
28	0.557399723374827	0.165283540802213	0.732885906040268	0.321022727272727	0.527054108216433	0.765258215962441	-5.74	22.26	-9.49	56.47	16.68	73.93407091267 24
29	0.506426735218509	0.028920308483290 6	0.697986577181208	0.234135667396061	0.426716141001855	0.728395061728395	1.59	43.48	-3.36	54.13	-4.889999999999999	64.74083913649 02
30	0.586360739324441	0.166347992351816	0.823121387283237	0.369146005509642	0.580270793036775	0.823754789272031	-0.19	43.08	-16.21	58.73	5.75	64.92888504753 67
31	0.450617283950617	0.061728395061728 5	0.725274725274725	0.251732101616628	0.367058823529412	0.755760368663594	-4.55	52.62	-12.21	64.54	13.65	70.97724346340 82
32	0.604570637119114	0.074792243767313 2	0.648714810281518	0.242138364779874	0.443531827515 4	0.777292576419214	-3.68	40.93	15.26	66.67	-17.55	71.30561068702 29
33	0.592485549132948	0.084537572254335 2	0.680263157894737	0.273311897106109	0.485900216919 74	0.748878923766816	2.59	21.36	0	39.07	-0.929999999999993	66.23166775244 3
34	0.632974910394265	0.112544802867384	0.594497607655502	0.291497975708502	0.421610169491525	0.715555555555556	3.2	24.53	-6.5	46.05	2.009999999999999	70.14649360535 93
35	0.697146562905318	0.058365758754863 9	0.563618290258449	0.187845303867403	0.429906542056075	0.672489082969432	-1.81	28.58	-10.01	43.6	10.63	69.63484060747 02
36	0.620051085568327	0.101532567049808	0.653508771929824	0.243076923076923	0.472727272727273	0.651639344262295	-0.73	23.1	-6.14	53.24	7.25	72.60996702637 89
37	0.691218130311615	0.094192634560907 1	0.676243093922652	0.206349206349206	0.506756756756757	0.775100401606426	2.07	34.05	8.72	47.26	-6.77	75.29009179265 66
38	0.642135642135642	0.069985569985570 5	0.608799048751486	0.217741935483871	0.453563714902808	0.708860759493671	-0.44	23.71	1.92	47.94	-5.7	72.83553562187 34
39	0.692762186115214	0.028064992614475 3	0.704880817253121	0.291390728476821	0.533769063180828	0.817757009345794	1.99	34.49	-8.08	41.9	-2.94	71.97081485215 06
40	0.5512256442489	0.057196731615336 1	0.702767749699158	0.316666666666667	0.494208494208494	0.748120300751 8	1.58	28.08	-15.52	52.17	6.460000000000001	69.46499165781 89
41	0.553010471204188	0.060209424083769 8	0.751592356687898	0.319512195121951	0.525616698292222	0.760330578512397	-1.06	27.87	-6.64	47.26	2.319999999999999	66.57012993039 44

42	0.4689603590127 15	0.0912490650710548	0.7702936096718 48	0.278959810874704	0.5104408352668 21	0.6834862385321 1	0.35	27.99	-1.66	48.25	-1.139999 99999999	63.85613305613 3
43	0.4551435406698 56	0.0448564593301436	0.7643835616438 36	0.342560553633218	0.4861111111111111	0.7401574803149 61	-1.68	31.36	4.87	45	-7.9	63.77
44	0.5811623246492 99	0.0661322645290583	0.6683168316831 68	0.367816091954023	0.4899193548387 1	0.7632653061224 49	0.48	27.28	5.64	37.88	-6.4	69.05
45	0.4980392156862 75	0.0732026143790851	0.7854137447405 33	0.289823008849558	0.4550898203592 81	0.828	-4.27	12	14.07	43.8	-5.05	69.60549930167 6
46	0.7401685393258 43	0.1516853932584 27	0.6024340770791 08	0.257425742574257	0.4521739130434 78	0.6926406926406 93	-4.72	37.69	7.91	42.46	-0.869999 99999999 7	70.75664012738 85
47	0.6765119549929 68	0.0492264416315048	0.6756453423120 09	0.298578199052133	0.4873949579831 93	0.7633928571428 57	-5.76	20	2.53	33.54	4.919999 99999999	58.96099732858 41
48	0.6931216931216 93	0.0252792475014697	0.6754785779398 36	0.3288888888888889	0.5152027027027 03	0.7977941176470 59	-4.94	35.71	7.91	46.45	-0.549999 99999999 7	64.53
49	0.5172881355932 2	0.0867796610169494	0.75	0.283410138248848	0.4563492063492 06	0.7596566523605 15	5.66	46.41	5.2	57.33	-7.95	71.55482162559 77
50	0.6809116809116 81	0.1303418803418 8	0.6996625421822 27	0.256038647342995	0.5249457700650 76	0.7754237288135 59	-1.54	17.65	4.86	47.81	-3.19	72.67232142857 14
51	0.4551083591331 27	0.0565015479876161	0.8175313059033 99	0.319634703196347	0.5093896713615 02	0.7934272300469 48	-3.79	19.23	3.96	47.43	0.579999 99999999 8	73.16090894670 05
52	0.6596491228070 18	0.0336842105263158	0.6662895927601 81	0.232227488151659	0.44	0.8578199052132 7	-1.46	0	-14.7	51.68	9.39	75.38
53	0.4641288433382 14	0.1149341142020 49	0.8105802047781 57	0.421383647798742	0.5271149674620 39	0.8385650224215 25	-4.02	37.94	-6.99	45.73	14.73	67.64782896444 33
54	0.53125	0.0922256097560 97	0.7417417417417 42	0.317142857142857	0.5140845070422 54	0.7837837837837 84	-3.78	25.93	-6.98	50.23	12.74	62.07954063604 24
55	0.5207308503162 33	0.0702740688685876	0.7784172661870 5	0.250614250614251	0.4512711864406 78	0.8111587982832 62	-2.89	25.71	-3.85	42.6	8.62	59.995019696117
56	0.6004382761139 52	0.1197954711468 22	0.6688227684346 7	0.300724637681159	0.5448113207547 17	0.7235772357723 58	4.66	32	-3.77	61.96	0.950000 00000000 3	63.63363124793 11
57	0.5209235209235 21	0.0750360750360749	0.7996987951807 23	0.345454545454545	0.5316742081447 96	0.8093220338983 05	2.02	15.06	-1.16	43.36	3.710000 00000001	73.02196547546 94
58	0.7767253044654 94	0.1447902571041 95	0.5837937384898 71	0.0972222222222222	0.4795321637426 9	0.6422413793103 45	2.63	25.27	-2.49	57.56	-0.030000 00000000 11	73.48520741671 9

59	0.570200573065903	0.057306590257879 7	0.696808510638298	0.179245283018868	0.424507658643326	0.808333333333333	-1.31	12.23	-2.61	47.15	3.56	71.598379279006 4
60	0.584474885844749	0.131659056316591	0.737288135593 2	0.326086956521739	0.524886877828054	0.849514563106796	-4.92	13.33	-9.59	42.15	9.180000000000001	73.826436999271 6
61	0.496158770806658	0.017285531370038 4	0.709766162310867	0.251612903225806	0.460854092526669	0.712328767123288	5.69	48.16	5.4	62.15	-6.48	72.67
62	0.615181518151815	0.069306930693069 3	0.693287037037037	0.333333333333333	0.502032520325203	0.806201550387597	1.25	51.84	1.6	55.7	-1.17	73.4782464
63	0.516591251885337	0.042232277526395 3	0.706539074960128	0.313725490196078	0.467441860465116	0.808612440191388	6.95	36.11	5.71	63.69	-13.43	70.462402610646 5
64	0.667151162790698	0.039244186046511 7	0.756662804171495	0.253807106598985	0.616883116883117	0.792626728110599	0.5	39.15	6.4	58.77	-7.23	74.169218989280 2
65	0.596125186289121	0.084202682563338 2	0.648575305291723	0.319727891156463	0.490486257928118	0.65625	-3.88	34.59	0.13	41.29	0.870000000000005	75.11
66	0.680515759312321	0.125358166189111	0.613895216400911	0.256410256410256	0.404612159329 14	0.758454106280193	-3.87	12.56	0.31	41.6	0.090000000000034	71.738927227107 16
67	0.570588235294118	0.048366013071895 6	0.578105781057811	0.220338983050847	0.388990825688073	0.663551401869159	0.42	47.89	6.56	65	-10.07	72.764155373162 8
68	0.532874139010645	0.063243581715716 7	0.686855670103093	0.211267605633803	0.399239543726236	0.713080168776371	-3.94	33.35	-0.16	49.67	5.900000000000001	67.370137677833 9
69	0.554298642533937	0.056561085972850 8	0.744047619047619	0.269841269841 27	0.468085106382979	0.791469194312796	-0.58	22.86	-1.12	52.9	5.520000000000001	76.882476050111 05
70	0.586723768736617	0.209850107066381	0.731382978723404	0.293159609120521	0.490697674418605	0.771929824561403	-2.9	8.11	-4.4	53.23	7.75	72.022695323434 88
71	0.687857142857143	0.025	0.736080178173719	0.233918128654971	0.530145530145 53	0.813084112149533	5.74	21.3	-6.7	45.18	-5.13	73.07
72	0.650069156293223	0.123789764868603	0.716743119266055	0.437751004016064	0.599137931034483	0.797468354430 38	6.48	22.82	-12.83	50.88	2.65	78.536722472636 36
73	0.598343685300207	0.078674948240165 3	0.736511919698871	0.295302013422819	0.538135593220339	0.766519823788546	-2.5	29.161811023622	-17.65	29.73	29.38	63.729369752480 6
74	0.523574947220267	0.116819141449683	0.758620689655172	0.236979166666667	0.468682505399568	0.772151898734177	-6.16	100	-21.52	27.59	30.94	60
75	0.377281947261663	0.038539553752535 5	0.728301886792453	0.375394321766562	0.451492537313433	0.718309859154 93	-5.96	33.33	-13.54	35.19	21.67	61.256688458338 3

76	0.6196189131968 95	0.0797459 42131263 1	0.5661764705882 35	0.2943396 22641509	0.3775933609958 51	0.75	9.47	34.02	-18.06	31.88	11.38	67.00023147494 71
77	0.4890604890604 89	0.0456885 45688545 7	0.7836676217765 04	0.3083333 33333333	0.4842519685039 37	0.8314176245210 73	1.37	15.38	20.55	49.35	-16.05	79.65159896283 49
78	0.7140740740740 74	0.0925925 92592592 2	0.5689084895259 1	0.2283464 56692913	0.4405286343612 33	0.6467889908256 88	-0.51	0	24.28	44.24	-18.66	76.58553654743 39
79	0.5809968847352 03	0.1066978 19314642	0.7250712250712 25	0.2595155 70934256	0.4637002341920 37	0.7401960784313 73	4.28	43.86	21.71	48.36	-24.11	82.16830247718 38
80	0.6438871473543 23	0.0137931 03448275 9	0.6728971962616 82	0.3396226 41509434	0.5037593984962 4	0.7607843137254 9	0.23	44.2	4.96	53.06	-10.82	73.52537307297 02
81	0.4523489932885 91	0.0765100 67114093 8	0.7468152866242 04	0.3872832 3699422	0.5223300970873 79	0.7477477477477 48	2.04	33.33	-3.85	46.22	11.13	72.97
82	0.5487804878048 78	0.0337159 25394548	0.6713881019830 03	0.3352601 15606936	0.4415011037527 59	0.6710526315789 47	-1.67	40.81	2.78	46.37	0.569999 99999999 3	78.2
83	0.6007246376811 6	0.0934782 60869565 3	0.7683397683397 68	0.3104693 14079422	0.5956043956043 96	0.7747747747747 75	-3.4	13.33	10.35	41.82	-5.25	73.13
84	0.6560463432295 44	0.0521361 33236785	0.6919431279620 85	0.2198275 86206897	0.4623430962343 1	0.7666666666666 67	-5.35	27.2	-3.37	44.6	11.3	73.91
85	0.4276901987662 78	0.1048663 46812885	0.8196147110332 75	0.3383742 91115312	0.4308943089430 89	0.8472222222222 22	-1.31	8.33	-7.85	42.77	11.73	67.52097218471 51
86	0.5825314581791 27	0.0843819 39304219 2	0.7103064066852 37	0.2409240 92409241	0.4331797235023 04	0.8037383177570 09	0.03	35.94	14.65	54.37	-12.79	69.94228577044 03
87	0.4739970282317 98	0.0839524 51708766 8	0.7576271186440 68	0.3302752 29357798	0.4825986078886 31	0.7209302325581 4	-5.08	41.57	3.14	61.33	-2.61	67.42451963241 44
88	0.5834532374100 72	0.1107913 66906475	0.7125506072874 5	0.3612903 22580645	0.5740318906605 92	0.7695652173913 04	1.2	22.64	12.38	53.84	-12.9	73.33
89	0.6177811550151 98	0.0683890 57750759 8	0.7343124165554 07	0.3886792 45283019	0.5219399538106 24	0.8476190476190 48	4.97	32.21	4.51	49	-14.15	74.43518472395 18
90	0.6846021751574 13	0.2112192 32970807	0.4906832298136 65	0.2102803 73831776	0.3264248704663 21	0.5827338129496 4	0.44	29.34	0.39	54.43	-9.44	70.07045915544 9
91	0.4865794907088 78	0.1940812 11286993	0.6977099236641 22	0.2725274 72527473	0.4059829059829 06	0.7478991596638 66	9.93	39.52	17.69	56.31	-28.74	76.14408362989 32
92	0.6375545851528 38	0.1812227 07423581	0.7354430379746 84	0.2938596 49122807	0.5745614035087 72	0.7980769230769 23	2.46	19.67	-12.54	45.37	10.29	75.52632310779 82

93	0.49323308270 6767	0.03082706766917267	0.7340946166394 78	0.28428927680798	0.48608137044967 9	0.76530612244898	-0.86	31.75	-1.62	47.84	2.68000000000001	66.10260575296 1
94	0.52671232876712 3	0.04726027397260 28	0.72587412587412 6	0.27360774818401 9	0.48747591522158	0.72123893805309 7	-5.4	44.71	-3.55	50	7.75	69.66523242789 13
95	0.61453744493392 1	0.06167400881057 25	0.66402116402116 4	0.29197080291970 8	0.44943820224719 1	0.72093023255814	-2.3	14.29	-10.21	45.77	12.33	71.19421045594 33
96	0.76676557863501 5	0.52818991097922 8	0.553261767134 6	0.27835051546391 8	0.42730496453900 7	0.71755725190839 7	-3.86	13.04	5.43	52.87	0.62999999999999 5	68.48718469217 97
97	0.59882005899705	0.08652900688298 92	0.753108348134 991	0.37668161434977 6	0.57647058823529 4	0.73619631901840 5	-1.71	22.64	0.01	47.58	7.22	72.82757009345 8
98	0.70920502092050 2	0.10146443514644 4	0.595879556259 905	0.28723404255319 1	0.43518518518518 5	0.75	2.58	33.33	-1.94	48.09	1.56999999999999	77.44
99	0.53599114064230 3	0.02879291251384 27	0.656950672645 74	0.26222222222222 2	0.44370860927152 3	0.70503597122302 2	-3.77	40.02	-3.28	44.06	8.04	74.71591489361 7
100	0.44053774560496 4	0.04860392967942 09	0.778061224489 796	0.23055555555555 6	0.36746987951808 72	0.69178082191780 8	-0.63	30	-5.1	44.78	6.76	69.3
101	0.54413191076624 6	0.04946653734238 58	0.714007782101 167	0.23465703971119 1	0.40356083086053 4	0.74213836477987 4	0.29	20.51	-4.32	40.77	9.65000000000001	63.16052236732 42
102	0.41941747572815 5	0.04951456310679 62	0.738035264483 627	0.33333333333333 3	0.47800586510263 9	0.67272727272727 3	-3.1	25.71	-3.26	35.88	8.90000000000001	67.41015708970 65
103	0.67420349434737 9	0.02877697841726 63	0.729372937293 729	0.33333333333333 3	0.60126582278481	0.80519480519480 5	-0.33	16.68	2.57	32.01	-2.59	60.87
104	0.55725971370143 1	0.07157464212678 94	0.753507014028 056	0.25619834710743 8	0.47826086956521 7	0.79166666666666 7	-0.13	18.43	-2.88	40.43	-1.17	70.86
105	0.50930460333006 9	0.08325171400587 66	0.759493670886 076	0.35191637630662	0.47740112994350 3	0.87234042553191 5	-2.87	19.05	0.21	49.29	5.39	64.71
106	0.53993933265925 2	0.10212335692618 8	0.774647887323 944	0.25882352941176 5	0.50310559006211 2	0.82098765432098 8	1.03	17.35	-7.48	48.24	7.27	62.05575133915 99
107	0.39640591966173 4	0.04545454545454 53	0.807909604519 774	0.32	0.453125	0.76510067114094	10.4	33.89	-0.72	47.88	-10.84	65.77643355288 65
108	0.53232758620689 7	0.07543103448275 86	0.615879828326 18	0.37549407114624 5	0.42492012779557	0.70198675496688 7	9.91	52.32	5.86	65.17	-25.13	68.24
109	0.52068617558022 2	0.04339051463168 51	0.761506276150 628	0.31972789115646 3	0.46439628482972 1	0.74534161490683 2	1.85	17.98	23.09	41.47	-19.82	73.93013173370 97

110	0.5673913043478260869	0.038043478260869	0.6639511201629339	0.36322869955157	0.5304878048780495	0.6842105263157959	4.79	26.54	22.69	47.14	-24.9	68.26
111	0.4960317460317460	0.059523809523809	0.7456521739130433	0.27124183006536	0.4698795180722289	0.6973684210526332	-0.8	23.23	13.18	46.77	-12.85	75.240068534569
112	0.6931589537223340	0.020120724346076	0.6823161189358375	0.24113475177305	0.4883720930232560	0.6887417218543005	2.54	38.46	13.9	51.4	-14.24	74.15
113	0.3456121343445290	0.075839653304442	0.8271186440677970	0.418491484184915	0.5015290519877680	0.7741935483870970	5.23	35.48	7.62	54.84	-8.59	74.15
114	0.5865580448065170	0.033604887983706	0.7024029574861370	0.30769230769230	0.4639498432601880	0.7076023391812860	3.6	36.72	-3.1	46.61	1.36	74.959097339689
115	0.5479910714285710	0.0546875	0.7719298245614040	0.31877729257641	0.5714285714285710	0.7651006711409400	-3.92	25.03	-10.67	38.42	14.05	73.854217749313
116	0.6048667439165170	0.033603707995365200	0.6417004048583	0.35838150289017	0.4599303135888500	0.7482014388489210	7.52	18.19	13.09	40.54	-20.56	68.506694937034
117	0.4876126126126130	0.074324324324323	0.7969924812030070	0.35714285714285	0.5187713310580200	0.7898550724637680	-2.25	19.61	-7.52	46.8	10.4	75.566438883541900
118	0.4446601941747570	0.023300970873786500	0.7809523809523810	0.27548209366391200	0.4355300859598500	0.7402597402597400	-2.53	20.51	-2.75	46.42	6.539999999999999	69.114053553624800
119	0.6680369989722510	0.195272353545735	0.6903973509933770	0.24836601307189500	0.4889589905362780	0.7371794871794870	-2.23	13.64	-5.72	38.67	9.29	69.168721951219500
120	0.6467341306347750	0.182152713891444	0.6128526645768020	0.30102040816326500	0.4827586206896550	0.7314285714285710	-2.6	18.52	-2.76	32.24	7.98	78.395925871015600
121	0.6057692307692310	0.074786324786324700	0.6086956521739130	0.30769230769230800	0.4713804713804710	0.6518987341772150	8.37	27.52	3.49	44.52	-7.999999999999999	65.677423154794800
122	0.6369565217391300	0.153260869565217	0.6275229357987980	0.19424460431654700	0.4567474048442910	0.6827586206896550	-1.36	26.53	5.82	42.51	-9.18	74.176388328133600
123	0.5389908256880730	0.090596330275229400	0.7291196388261850	0.29302325581395300	0.4982456140350880	0.7851851851851850	-1.4	36	-11	49.19	6.139999999999999	69.07
124	0.3730853391684900	0.060175054704595	0.7278481012658230	0.30026809651474500	0.4316770186335400	0.6811594202898550	-5.72	44.44	-15.18	49.35	18.63	72.95
125	0.4458293384467880	0.033557046979865900	0.7909090909090910	0.32460732984293200	0.5190217391304350	0.7333333333333330	0.63	19.28	-3.29	36.69	7.320000000000001	65.111293610911700
126	0.5373592630501540	0.048106448311156800	0.7567010309278350	0.25	0.4490445859872610	0.7055214723926380	-2.48	55	-6.26	55.05	9.44	73.68

127	0.6178067318132 47	0.0977198697068404	0.7820267686424 47	0.2891566 26506024	0.5444839857651 24	0.7861635220125 79	0.11	40	1.67	47.79	-2.88	67.53167673716 01
128	0.7742946708463 95	0.5673981 19122257	0.5513196480938 42	0.25	0.3980582524271 84	0.7071428571428 57	-2.28	40.03	-4.19	50.67	-2.13	71.68909312870 6

Dataset 2: PCA and Cluster Dataset

	PC1	PC2	PC3	Cluster	Team	Year	OC	Pts.Poss	Result	Promotion	Fired	OC_initials
1	-1.10323584786854	2.0989056987752	1.24185329764684	1	Arizona Cardinals	2022	Kliff Kingsbury	1.9	OC	Demoted	Yes	KK
2	-2.97049743010895	1.88817027109315	-1.8484341330857	1	Arizona Cardinals	2023	Drew Petzing	2.54	active	Same	No	DP
3	-3.02811790392434	0.17118678442 8119	0.4152955180888 34	1	Arizona Cardinals	2024	Drew Petzing	2.44	active	Same	No	DP
4	-0.36472319510 0122	-0.7527311012 0034	0.7933776446865 49	3	Arizona Cardinals	2025	Drew Petzing	2.26	active	Same	No	DP
5	-1.89513460831879	-1.9375583010406	-1.7941587455 0416	3	Atlanta Falcons	2022	Dave Ragone	2.17	active	Same	No	DR
6	-5.2210383734653	-2.55001031137062	-3.1385463427 8629	3	Atlanta Falcons	2023	Dave Ragone	1.95	Position Coach	Demoted	Yes	DR
7	3.64931042851924	2.79878240116 4	0.6626088014055 36	2	Atlanta Falcons	2024	Zach Robinson	2.39	active	Same	No	ZR
8	-1.07529436378974	0.0799269352891894	-0.6398150324 50388	3	Atlanta Falcons	2025	Zach Robinson	1.98	active	Same	No	ZR
9	-4.48918682210103	-5.24715044779756	1.0903146115662 4	3	Baltimore Ravens	2022	Greg Roman	2.08	OC	Same	No	GR
10	0.2800489460822942	-1.5561011751 2335	1.3955965985320 9	3	Baltimore Ravens	2023	Todd Monken	2.67	active	Same	No	TM
11	-1.17328783028943	-3.5869084478 4073	0.3490307409926 26	3	Baltimore Ravens	2024	Todd Monken	2.78	active	Same	No	TM
12	-1.4001181825 7104	-2.5555358987 2356	1.1464774866564 9	3	Baltimore Ravens	2025	Todd Monken	2.36	active	Same	No	TM
13	1.72868736808544	-0.8019832323 95551	0.5089210880720 16	2	Buffalo Bills	2022	Ken Dorsey	2.68	active	Same	Np	KD
14	0.2541840870 3331	0.6075631973 83207	-0.7038643957 78374	2	Buffalo Bills	2023	Joe Brady	2.62	active	Same	No	JB
15	0.2541840870 3331	0.6075631973 83207	-0.7038643957 78374	2	Buffalo Bills	2023	Joe Brady	2.62	OC	Same	Yes	JB
16	0.2938962662 67777	0.5105816416 05459	-0.0363528714 798485	2	Buffalo Bills	2024	Joe Brady	2.73	active	Same	No	JB
17	0.5477966147 49701	-0.4357969960 89774	1.1162869737 2155	2	Buffalo Bills	2025	Joe Brady	2.89	active	Same	No	JB

18	1.93822459855749	-0.417891821056253	-0.967372655507213	2	Carolina Panthers	2022	Ben Mcadoo	1.79	assistant	Demoted	No	BM
19	3.69898471118749	0.803639780403576	-0.388271329465851	2	Carolina Panthers	2023	Thomas Brown	2.4	Position Coach	Demoted	No	TB
20	2.59447780340143	-0.383277431139717	-0.692821830499699	2	Carolina Panthers	2024	Brad Idzik	1.93	active	Same	No	BI
21	2.47344142153703	-0.494977157661578	0.21396456476953	2	Carolina Panthers	2025	Brad Idzik	2.08	active	Same	No	BI
22	1.60075820241405	-3.82462122264882	0.480146947098425	3	Chicago Bears	2022	Luke Getsy	1.82	active	Same	No	LG
23	0.05247364532 87684	-1.09853707998488	-0.103915659789437	3	Chicago Bears	2023	Luke Getsy	1.8	OC	Same	Yes	LG
24	0.98814206684 1936	0.926502737763524	-0.339669498555974	2	Chicago Bears	2024	Shane Waldron	1.86	Position Coach	Demoted	Yes	SW
25	0.98814206684 1936	0.926502737763524	-0.339669498555974	2	Chicago Bears	2024	Shane Waldron	1.86	Position Coach	Demoted	Yes	SW
26	-0.19417754620 5703	0.737043624769489	-1.44046338525236	2	Chicago Bears	2025	Declan Doyle	2.51	active	Same	No	DD
27	3.15480665859 96	1.421810050952 65	0.8154675504682 92	2	Cincinnati Bengals	2022	Brian Callahan	2.6	NA	NA	NA	BC
28	-2.76483089694 52	4.695643992432 29	0.629247730803 754	1	Cincinnati Bengals	2023	Brian Callahan	2.06	NA	NA	NA	BC
29	-0.70134527648 2773	0.231439102284614	-0.954778212888166	3	Cincinnati Bengals	2024	Dan Pitcher	2.79	NA	NA	NA	DP
30	1.27231007824 422	0.835679358321037	0.717058878314 166	2	Cincinnati Bengals	2025	Dan Pitcher	2.33	NA	NA	NA	DP
31	-0.55497491311 3184	-0.274001722467044	-2.3048550624704	3	Cleveland Browns	2022	Alex Van Pelt	2.23	active	Same	No	AVP
32	2.24549048442 727	-0.566120564330 67	0.130166867524 424	2	Cleveland Browns	2023	Alex Van Pelt	1.82	Position Coach	Demoted	Yes	AVP
33	-0.053165598610215	0.419371174323251	-3.09325490261 984	2	Cleveland Browns	2024	Ken Dorsey	1.42	Position Coach	Demoted	Yes	KD
34	-2.01154196929 552	-0.851080520188653	-0.461533595203983	3	Cleveland Browns	2025	Tommy Rees	1.56	active	Same	Np	TR
35	0.08351565836 67393	0.468429072525935	0.232285905099 354	2	Dallas Cowboys	2022	Kellen Moore	2.5	active	Same	No	KM
36	-1.02324787590 051	1.154863747278 04	-0.119183180985 652	1	Dallas Cowboys	2023	Brian Schottenheimer	2.48	active	Same	No	BS

37	-1.4065011103198	2.87849331202098	-0.61926118777916	1	Dallas Cowboys	2024	Brian Schottenheimer	2.2	Head Coach	Promoted	No	BS
38	-1.095980276581324	1.406570109314	-0.025822582444936 9	1	Dallas Cowboys	2025	Klayton Adams	2.78	active	Same	No	KA
39	-1.3025486496698 1	0.288377235355923	1.26201738405426	1	Denver Broncos	2022	Justin Outten	1.41	NA	NA	NA	JO
40	-1.5924972790163 8	1.019258925650 32	0.269951858446375	1	Denver Broncos	2023	Joe Lombardi	2.06	NA	NA	NA	JL
41	0.950791978077919	0.102064952402239	0.871982627939524	2	Denver Broncos	2024	Joe Lombardi	2.24	NA	NA	NA	JL
42	0.755137797273028	0.231949305022808	-0.362414516789958	2	Denver Broncos	2025	Joe Lombardi	2.25	NA	NA	NA	JL
43	1.247773611891 04	-0.133377774892752	-0.111350206381839	2	Detroit Lions	2022	Ben Johnson	2.83	NA	NA	NA	BJ
44	0.534310013375 84	-0.359871639305 13	-1.092269725873 98	2	Detroit Lions	2023	Ben Johnson	2.28	NA	NA	NA	BJ
45	0.653611114372582	-1.135806213653 04	-0.926453502625554	2	Detroit Lions	2024	Ben Johnson	3.13	NA	NA	NA	BJ
46	0.347988394775108	-0.470279548582404	0.48037101196187	2	Detroit Lions	2025	John Morton	3.05	NA	NA	NA	JM
47	1.232014370670 91	-0.932692969106141	0.262425286129059	2	Green Bay Packers	2022	Adam Stenavich	2.27	active	Same	No	AS
48	-1.767654323081 53	1.890775212432	0.397851450712391	1	Green Bay Packers	2023	Adam Stenavich	1.73	active	Same	No	AS
49	1.026770623630 05	1.464078067610 43	-0.027805278823 0374	2	Green Bay Packers	2024	Adam Stenavich	2.51	active	Same	No	AS
50	0.551353773250665	0.103054584440 88	0.207469339131601	2	Green Bay Packers	2025	Adam Stenavich	2.66	active	Same	No	AS
51	-0.832183600697315	-1.518447959993 91	-0.85754746370816	3	Houston Texans	2022	Pep Hamilton	1.52	assistant	Demoted	Yes	PH
52	-0.183847477836232	0.543248581081322	1.73112234940562	2	Houston Texans	2023	Bobby Slowik	1.98	active	Same	No	BS
53	1.726853409930 52	-1.137512445376 97	0.270905362825377	2	Houston Texans	2024	Bobby Slowik	1.83	Position Coach	Demoted	Yes	BS
54	0.643030018956405	1.473845750642 64	1.42669855272914	2	Houston Texans	2025	Nick Caley	2.13	active	Same	No	NC
55	2.952196382890 5	-0.701779250731223	-0.230427175341252	2	Indianapolis Colts	2022	Marcus Brady	1.48	Position Coach	Demoted	Yes	MB
56	1.917776505117 1	0.369417562430396	-0.729992559576137	2	Indianapolis Colts	2023	Jim Bob Carter	2.21	active	Same	No	JBC

57	1.754366701271106	0.537219605822506	-1.371416131760028	2	Indianapolis Colts	2024	Jim Bob Carter	2.18	active	Same	No	JBC
58	-0.464706041416739	-0.242351441580865	-0.060954723138569	3	Indianapolis Colts	2025	Jim Bob Carter	2.98	active	Same	No	JBC
59	1.931725383581143	-0.985921822644252	1.1449035978609	2	Jacksonville Jaguars	2022	Press Taylor	2.43	active	Same	No	PT
60	-3.445997338914833	2.603002794003440	0.8908117345418670	1	Jacksonville Jaguars	2023	Press Taylor	1.51	active	Same	No	PT
61	0.157472474287078	0.964688265941104	-0.031916077400 07923	2	Jacksonville Jaguars	2024	Press Taylor	2.06	Position Coach	Demoted	Yes	PT
62	2.387160757873490	0.671823024148723	1.4441432352818 8	2	Jacksonville Jaguars	2025	Grant Udinski	2.47	active	Same	No	GU
63	-1.318489996011830	-1.567320935065990	-1.536052418742880	3	Kansas City Chiefs	2022	Eric Bieniemy	2.76	OC	Same	No	EB
64	-0.081226015394281	-0.869631721209099	-0.026085760178 8527	3	Kansas City Chiefs	2023	Matt Nagy	1.94	active	Same	No	MN
65	-0.690967692995485	-2.253670928732590	-0.387710520622355	3	Kansas City Chiefs	2024	Matt Nagy	2.32	active	Same	No	MN
66	0.126553068766438	-1.129590172124930	1.4089018064361 6	3	Kansas City Chiefs	2025	Matt Nagy	2.6	active	Same	No	MN
67	-0.436639743736951	0.765449170976017	-0.003785795245 73682	2	Las Vegas Raiders	2022	Mick Lombardi	2.28	active	Same	No	ML
68	-1.158341920454360	1.984173915969960	1.0243438451175 1	1	Las Vegas Raiders	2023	Bo Hardegree	2.04	Position Coach	Demoted	Yes	BH
69	-1.158341920454360	1.984173915969960	1.0243438451175 1	1	Las Vegas Raiders	2023	Bo Hardegree	2.04	assistant	Demoted	Yes	BH
70	-3.249606890736700	0.018439853462268 2	-1.803927443195 75	1	Las Vegas Raiders	2024	Scott Turner	1.86	Position Coach	Demoted	Yes	ST
71	-3.249606890736700	0.018439853462268 2	-1.803927443195 75	1	Las Vegas Raiders	2024	Scott Turner	1.86	assistant	Demoted	Yes	ST
72	-0.988466167937623	1.202313811289 46	-1.5188110318258 1	1	Las Vegas Raiders	2025	Chip Kelley	1.65	active	Same	No	CK
73	0.408392095772609	-0.182912892216417	0.4240090621262 38	2	Los Angeles Chargers	2022	Kellen Moore	2.2	NA	NA	NA	KM

74	0.362441559210599	0.970699655428886	1.48147670855485	2	Los Angeles Chargers	2023	Kellen Moore	1.8	NA	NA	NA	KM
75	0.484463158605888	-0.0832909501505382	1.37079099494437	2	Los Angeles Chargers	2024	Greg Roman	2.2	NA	NA	NA	GR
76	1.22439608996618	-0.880547004448338	2.82164717643717	2	Los Angeles Chargers	2025	Greg Roman	2.35	NA	NA	NA	GR
77	2.92419666276668	2.10905244346011	-0.393986891679593	2	Los Angeles Rams	2022	Liam Coen	1.86	OC	Same	No	LC
78	2.14500399661717	2.53015603695443	-4.22736473522442	2	Los Angeles Rams	2023	Mike LaFleur	2.28	active	Same	No	ML
79	2.10935634151881	0.940031949561992	-2.19203378559924	2	Los Angeles Rams	2024	Mike LaFleur	2.31	active	Same	No	ML
80	-0.423238036989911	1.78910734321782	-1.024064261955877	1	Los Angeles Rams	2025	Mike LaFleur	3.16	active	Same	No	ML
81	0.240032708870784	-2.52358418775036	1.40355710070609	3	Miami Dolphins	2022	Frank Smith	2.31	active	Same	No	FS
82	-3.312498737150801734	0.709931955101734	2.12204878595814	1	Miami Dolphins	2023	Frank Smith	2.16	active	Same	No	FS
83	-2.189145093633233	-1.802246316370008	0.8530488736308893	3	Miami Dolphins	2024	Frank Smith	2.01	active	Same	No	FS
84	-0.588481105754675	-1.009190051615505	0.2384668697572218	3	Miami Dolphins	2025	Frank Smith	2.21	active	Same	No	FS
85	1.395557691655508	-0.919765480177044	-0.100327549012785	2	Minnesota Vikings	2022	Wes Phillips	2.23	active	Same	No	WP
86	-0.694284833881164	-0.187202354088264	-0.674194203267436	3	Minnesota Vikings	2023	Wes Phillips	2.07	active	Same	No	WP
87	1.250059388152971	-0.608528437971259	2.12838731821447	2	Minnesota Vikings	2024	Wes Phillips	2.32	active	Same	No	WP
88	0.167337165130593	1.654094414933862	0.2244339690139822	2	Minnesota Vikings	2025	Wes Phillips	1.81	active	Same	No	WP
89	2.761100543556449	-0.296272169999547	-0.375931960388085	2	New England Patriots	2022	Matt Patrcia	1.89	assistant	Demoted	Yes	MP
90	-1.148877303299170	-0.834379131310137	-0.267676225515527	3	New England Patriots	2023	Bill O'Brien	2.19	College OC	Demoted	Yes	BO'Brien
91	0.131913732349923	-0.966838837582287	-1.378061954219840	3	New England Patriots	2024	Alex Van Pelt	1.74	assistant	Demoted	Yes	AVP
92	-0.082713967471084 7	-1.640105247135020	1.8806870311456 1	3	New England Patriots	2025	Josh Mcdaniels	2.7	active	Same	No	JM

93	0.629788861061373	-2.006661038786636	1.44895632988026	3	New Orleans Saints	2022	Pete Carmichael Jr	1.81	active	Same	No	PCJ
94	-4.746786054046663	2.604424402505534	-0.773983312728571	1	New Orleans Saints	2023	Pete Carmichael Jr	2.58	assistant	Demoted	Yes	PCJ
95	-2.643525628935	-2.309815708142 16	-0.142589669423 75	3	New Orleans Saints	2024	Klint Kubiak	1.9	OC	Same	No	KK
96	1.413271648365 88	0.741239854632376	2.030433160313 85	2	New Orleans Saints	2025	Doug Nussmeier	1.7	active	Same	No	DN
97	0.956965403307018	-0.377825683888464	-1.092586516825 72	2	New York Giants	2022	Mike Kafta	2.12	active	Same	No	MK
98	0.523517071401778	0.396902452355796	-1.389161123544 59	2	New York Giants	2023	Mike Kafta	1.73	active	Same	No	MK
99	0.416695709739819	1.469650772529 82	0.058575534991 3722	2	New York Giants	2024	Mike Kafta	1.67	active	Same	No	MK
100	-3.223256685695 42	3.509643357786 01	2.773593300375 68	1	New York Giants	2025	Mike Kafta	2.36	active	Same	No	MK
101	1.471897367526 17	-0.416511565560334	1.243755540010 64	2	New York Jets	2022	Mike LaFleur	1.59	OC	Same	No	ML
102	-1.462793289718 98	0.994764935177392	0.814033116561 909	1	New York Jets	2023	Nathaniel Hackett	1.62	active	Same	No	NH
103	-0.342583015024376	0.917622496493691	-1.044087716702 92	2	New York Jets	2024	Nathaniel Hackett	2	assistant	Demoted	Yes	NH
104	-0.105578699080912	0.490771021359206	-2.189388103417 22	2	New York Jets	2025	Tanner Engstrand	1.95	active	Same	No	TE
105	-0.079523450641725	1.277101926236 73	-1.016611774407 38	2	Philadelphia Eagles	2022	Shane Steichen	2.77	Head Coach	Promoted	No	SS
106	1.355342267448 63	0.174722984632987	-1.333774459637 18	2	Philadelphia Eagles	2023	Brian Johnson	1.77	Position Coach	Demoted	Yes	BJ
107	2.147157200627 7	-0.062288694228 7582	1.369878075738 63	2	Philadelphia Eagles	2024	Kellen Moore	2.5	Head Coach	Promoted	No	KM
108	0.913899187081142	0.179866403309112	0.364312014227 564	2	Philadelphia Eagles	2025	Kevin Patullo	2.14	active	Same	No	KP
109	1.979277526250 31	-0.605849583443184	0.070888604620 9166	2	Pittsburgh Steelers	2022	Matt Canada	1.79	active	Demoted	No	MC
110	1.817546161173 03	0.173902652868683	-0.292919468150 89	2	Pittsburgh Steelers	2023	Eddie Faulkner	1.86	Out of Work	Demoted	Yes	EF

111	1.81754616117303	0.173902652868683	-0.292919468150089	2	Pittsburgh Steelers	2023	Eddie Faulkner	1.86	Out of Work	Demoted	Yes	EF
112	0.361454951799422	-2.31152190008492	-1.29804445542519	3	Pittsburgh Steelers	2024	Arthur Smith	2.07	active	Same	No	AS
113	-2.8992034395273	-2.28415340953281	-1.1562184735069	3	Pittsburgh Steelers	2025	Arthur Smith	2.43	active	Same	No	AS
114	-0.510336562773068	-2.0204337664883	0.784219130277239	3	San Francisco 49ers	2022	Kyle Shanahan	2.63	active	Same	No	KS
115	-1.780423763312268	-2.08765852581963	0.996867058463348	3	San Francisco 49ers	2023	Kyle Shanahan	2.41	active	Same	No	KS
116	-0.92907906974653	-0.974669334945633	0.239282255525701	3	San Francisco 49ers	2024	Kyle Shanahan	2.34	active	Same	No	KS
117	-2.614994518888928	-0.340844856186678	0.808953759966984	1	San Francisco 49ers	2025	Klay Kubiak	2.48	active	Same	No	KK
118	0.779568871339905	-3.34324785699031	-0.203978708190935	3	Seattle Seahawks	2022	Shane Waldron	2.29	active	Same	No	SW
119	-0.419183475954572	-0.191543311243156	-0.249494255701666	3	Seattle Seahawks	2023	Shane Waldron	1.7	OC	Same	No	SW
120	2.512919708177567	0.468628062472224	0.656881321771288	2	Seattle Seahawks	2024	Ryan Grubb	2.13	College OC	Demoted	Yes	RG
121	-1.05558483400513	-1.43437178496114	0.855305238905039	3	Seattle Seahawks	2025	Klint Kubiak	2.58	active	Same	No	KK
122	2.22810394937258	-0.523660993708695	0.5701564051613	2	Tampa Bay Buccaneers	2022	Byron Leftwich	1.78	assistant	Demoted	Yes	BL
123	0.829539694012735	-0.06129858171313 63	-1.05186628436973	2	Tampa Bay Buccaneers	2023	Dave Canales	2.28	Head Coach	Promoted	No	DC
124	0.301403542460341	2.15235104837518	1.12125817825894	2	Tampa Bay Buccaneers	2024	Liam Coen	2.77	Head Coach	Promoted	No	LC
125	-0.263947438464167	2.02333889344995	1.85007460024985	1	Tampa Bay Buccaneers	2025	Josh Grizzard	2.25	active	Same	No	JG
126	-1.713434536896 48	-0.083601538691 3614	-0.194553712906161	3	Tennessee Titans	2022	Todd Downing	1.64	Position Coach	Demoted	Yes	TD

127	-1.98710969014318	1.36071343669706	0.540439643572333	1	Tennessee Titans	2023	Tim Kelly	1.84	Position Coach	Demoted	Yes	TK
128	1.21488903717584	0.25100674197162	-0.614352141197467	2	Tennessee Titans	2024	Nick Holz	1.9	active	Same	No	NH
129	1.01009883362465	0.682980347880138	-2.6253653842259	2	Tennessee Titans	2025	Nick Holz	1.52	active	Same	No	NH
130	1.98500513844194	-0.426617375412095	-0.528780330648421	2	Washington Commanders	2022	Scott Turner	1.72	Position Coach	Demoted	Yes	ST
131	-0.151852572582389	0.270455938586998	-1.86359922635938	2	Washington Commanders	2023	Eric Bieniemy	1.84	Position Coach	Demoted	Yes	EB
132	0.81195125885802	-0.494102042337015	0.237196724642513	2	Washington Commanders	2024	Kliff Kingsbury	2.76	active	Same	No	KK
133	-3.60997098054555	3.85313372824795	1.63067708799012	1	Washington Commanders	2025	Kliff Kingsbury	1.49	active	Same	No	KK

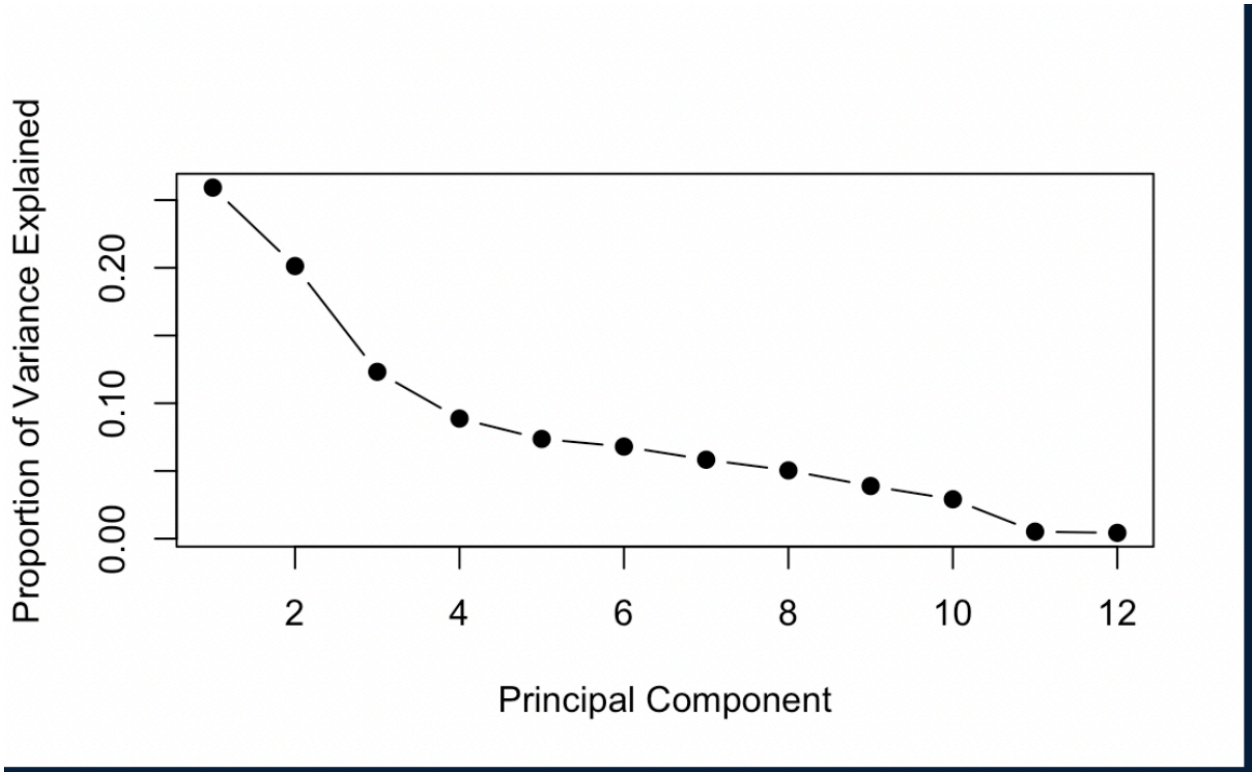


Figure 11: Principal Component Scree plot

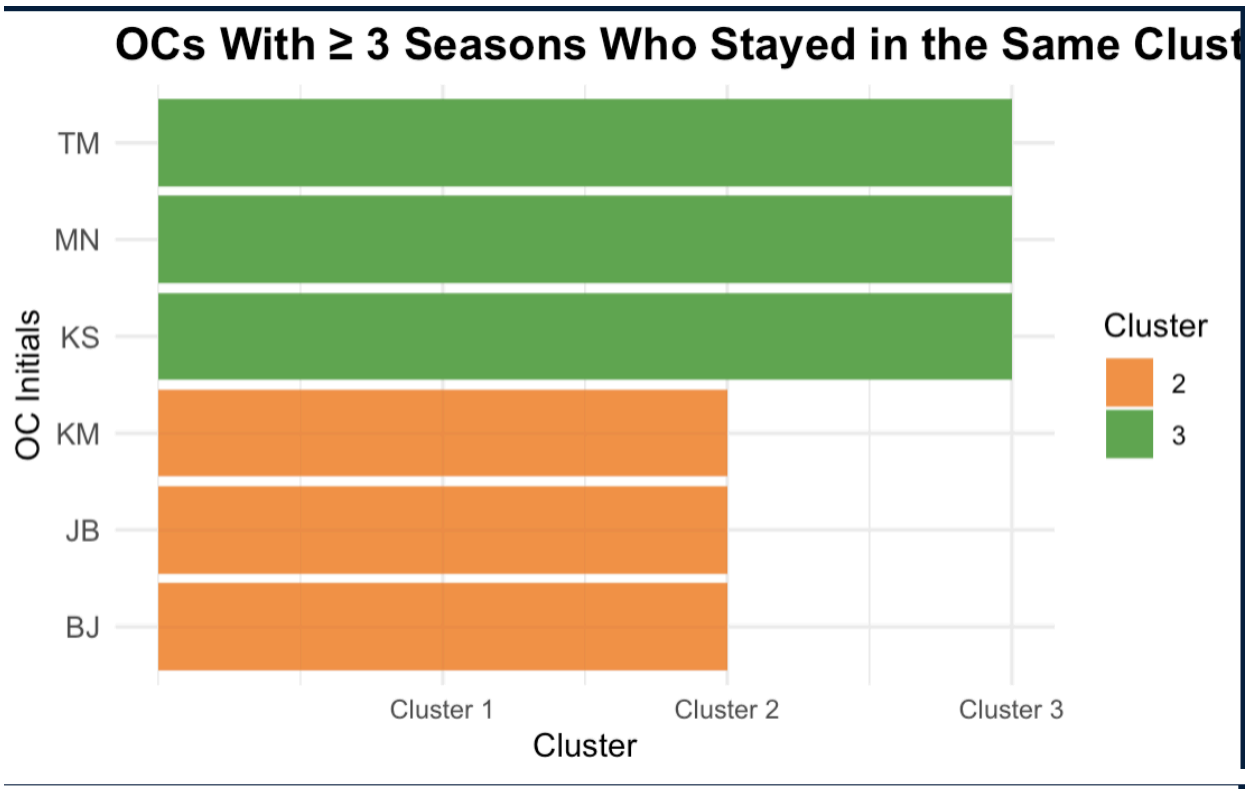


Figure 12: OC Cluster Stability

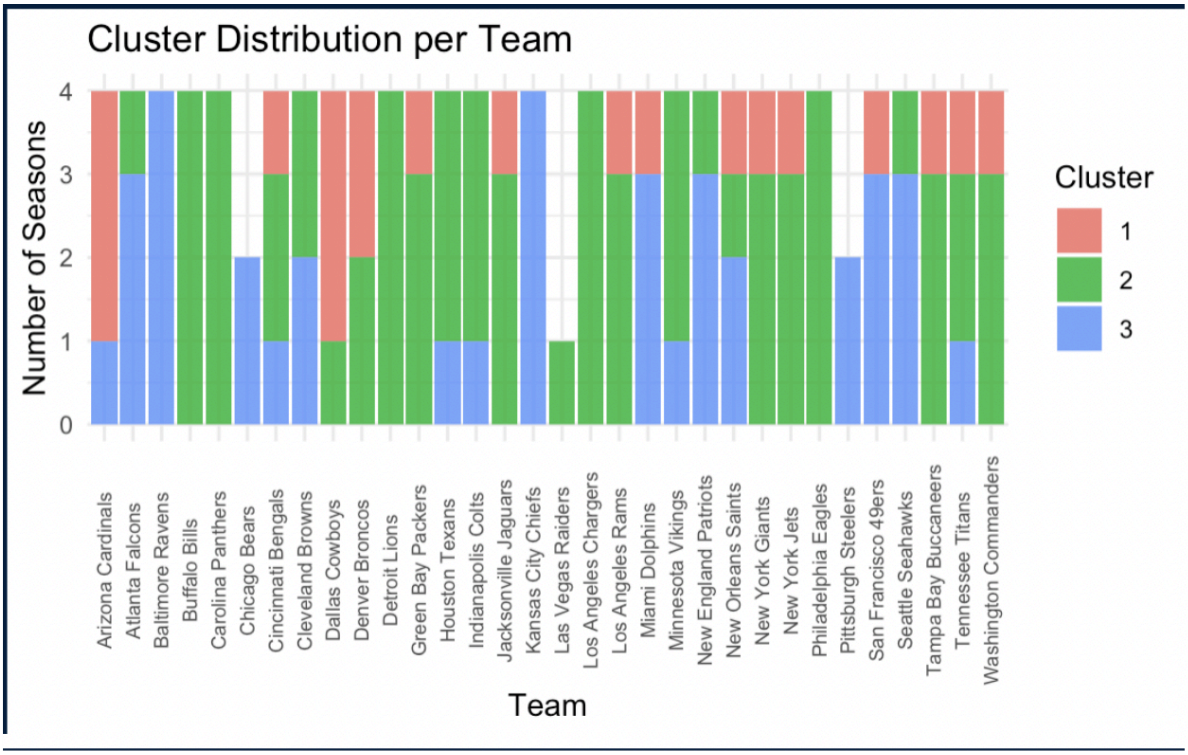


Figure 13: Cluster Distribution by Team

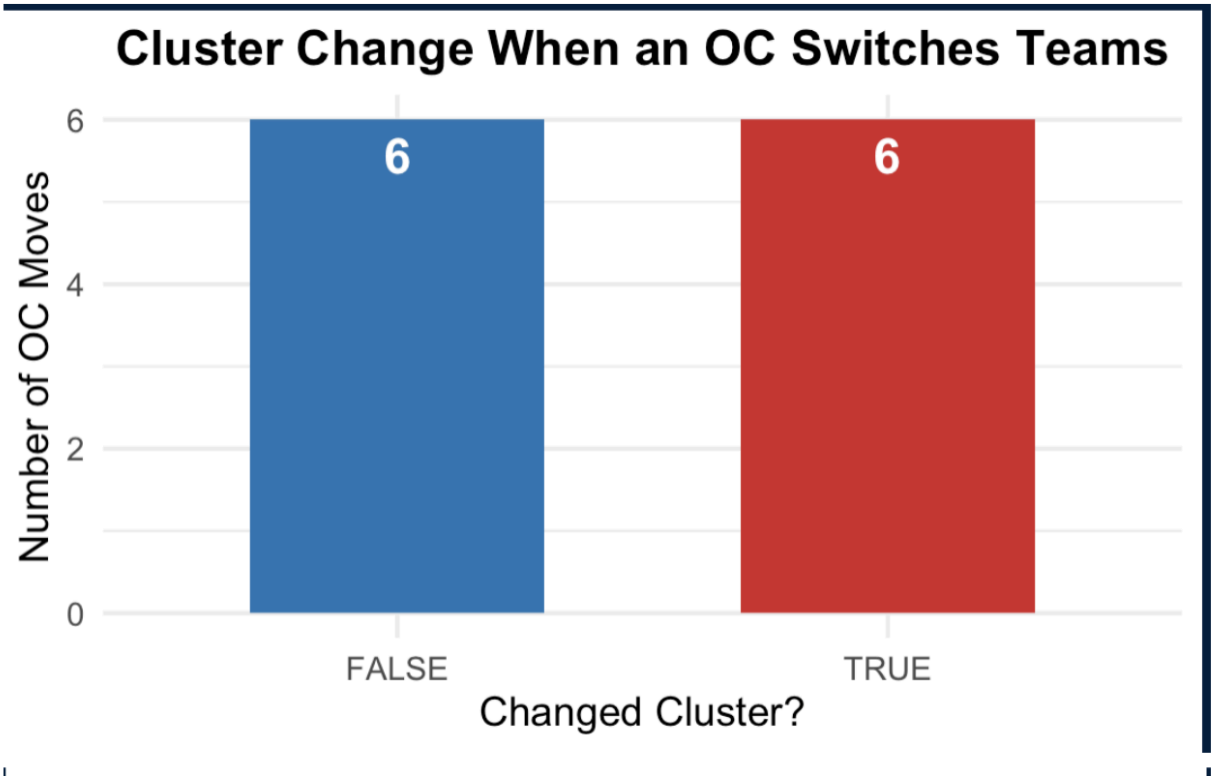


Figure 14: Offensive Coordinator New Team Scheme Changes

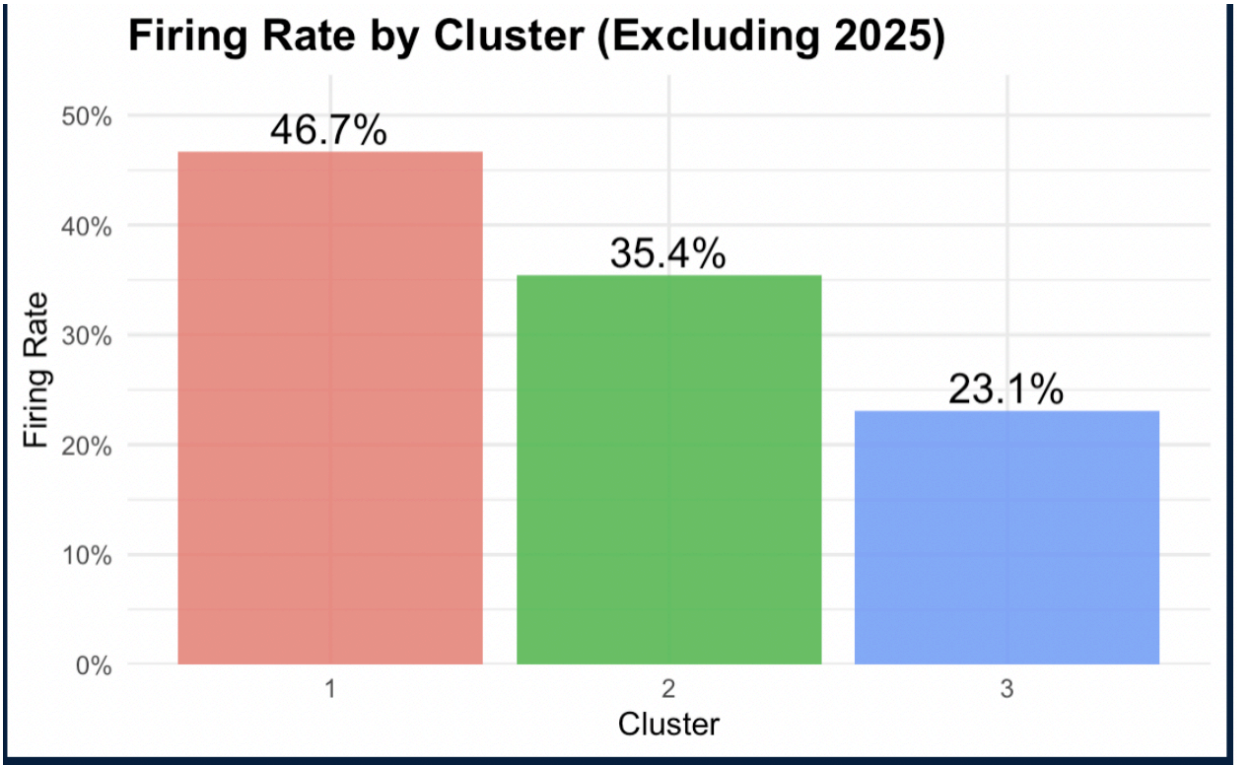
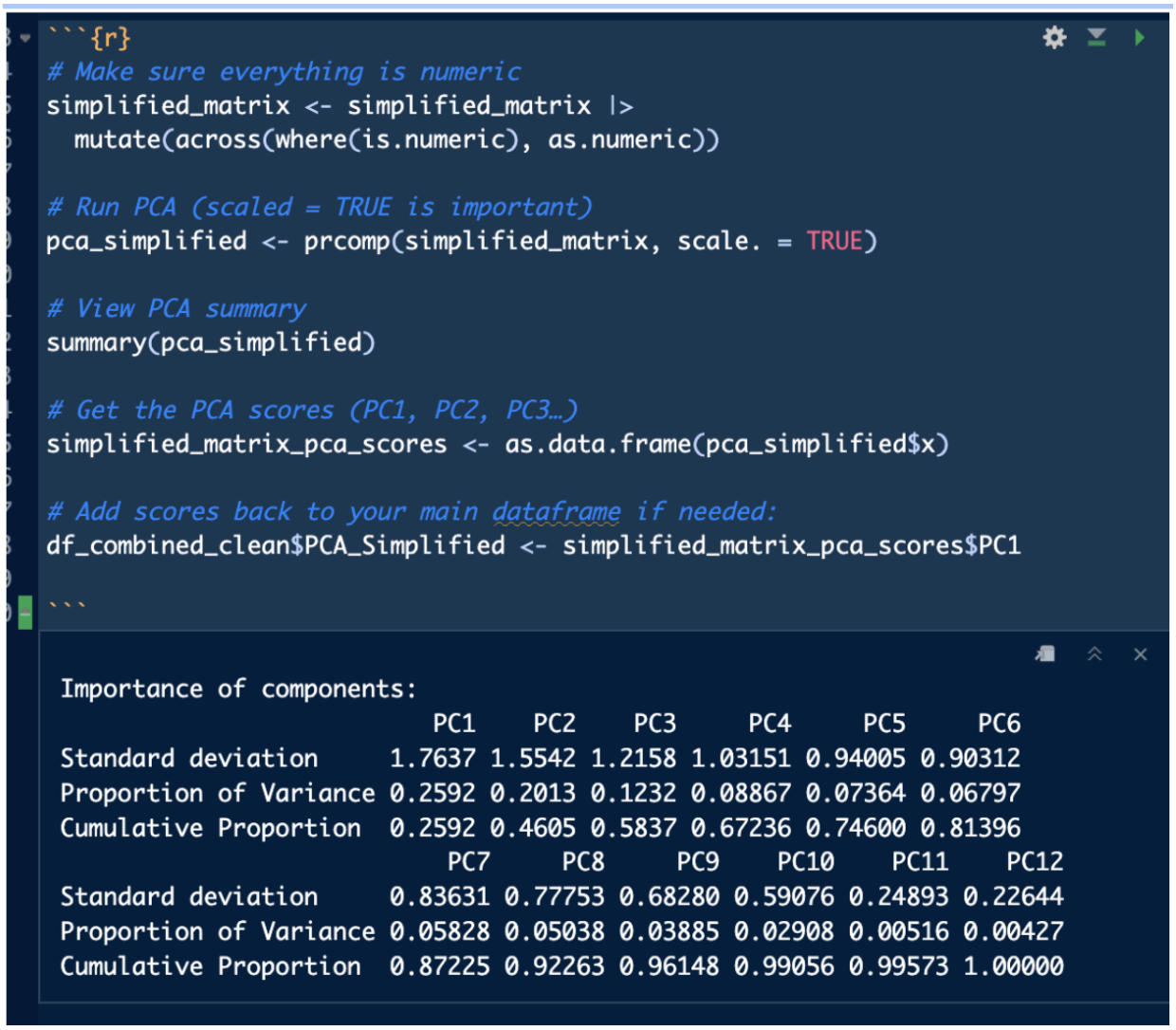


Figure 15: Firing Rate by Cluster



```r
# Make sure everything is numeric
simplified_matrix <- simplified_matrix |>
  mutate(across(where(is.numeric), as.numeric))

# Run PCA (scaled = TRUE is important)
pca_simplified <- prcomp(simplified_matrix, scale. = TRUE)

# View PCA summary
summary(pca_simplified)

# Get the PCA scores (PC1, PC2, PC3…)
simplified_matrix_pca_scores <- as.data.frame(pca_simplified$x)

# Add scores back to your main dataframe if needed:
df_combined_clean$PCA_Simplified <- simplified_matrix_pca_scores$PC1
```

```
Importance of components:
                          PC1    PC2    PC3     PC4     PC5     PC6
Standard deviation     1.7637 1.5542 1.2158 1.03151 0.94005 0.90312
Proportion of Variance 0.2592 0.2013 0.1232 0.08867 0.07364 0.06797
Cumulative Proportion  0.2592 0.4605 0.5837 0.67236 0.74600 0.81396
                          PC7     PC8     PC9    PC10    PC11    PC12
Standard deviation     0.83631 0.77753 0.68280 0.59076 0.24893 0.22644
Proportion of Variance 0.05828 0.05038 0.03885 0.02908 0.00516 0.00427
Cumulative Proportion  0.87225 0.92263 0.96148 0.99056 0.99573 1.00000
```

Coding Chunk 1: Importance of Components

```r
loadings <- pca_simplified$rotation
round(loadings, 3)
```

```
                          PC1    PC2    PC3    PC4    PC5    PC6    PC7
pct_shotgun            -0.210 -0.364  0.466 -0.034  0.162 -0.312 -0.339
pct_no_huddle          -0.149 -0.261  0.246 -0.384  0.517  0.456  0.318
pct_pass_shotgun        0.437  0.310 -0.069 -0.109 -0.111  0.238 -0.119
pct_pass_under_center   0.262  0.354  0.108 -0.080  0.346 -0.186  0.527
pct_pass_first          0.330  0.155  0.404 -0.132  0.205 -0.348 -0.327
pct_3rd_pass            0.385  0.165  0.231 -0.171  0.026  0.206 -0.269
0_1_WR_league_diff     -0.254  0.365  0.114  0.122  0.076 -0.397  0.275
0_1_WR_pass_rate       -0.139  0.082 -0.502 -0.484  0.185 -0.368 -0.189
2_WR_league_diff       -0.314  0.374  0.135  0.239  0.100  0.345 -0.267
2_WR_pass_rate         -0.279  0.237 -0.075 -0.582 -0.093  0.136 -0.183
3_4_WR_league_diff      0.372 -0.430 -0.123 -0.153 -0.138 -0.082  0.132
3_4_WR_pass_rate       -0.138  0.072  0.430 -0.344 -0.674 -0.055  0.281
                          PC8    PC9   PC10   PC11   PC12
pct_shotgun             0.045 -0.113 -0.194  0.305 -0.470
pct_no_huddle           0.037 -0.082  0.338 -0.037  0.047
pct_pass_shotgun        0.031 -0.007  0.428  0.304 -0.580
pct_pass_under_center  -0.280  0.023 -0.482  0.127 -0.167
pct_pass_first         -0.245  0.334  0.328 -0.180  0.323
pct_3rd_pass            0.451 -0.488 -0.343 -0.135  0.223
0_1_WR_league_diff      0.597 -0.061  0.315  0.245  0.127
0_1_WR_pass_rate       -0.219 -0.467  0.118  0.016  0.032
2_WR_league_diff       -0.363 -0.158 -0.028  0.489  0.297
2_WR_pass_rate          0.225  0.562 -0.291  0.105 -0.007
3_4_WR_league_diff      0.046  0.083 -0.023  0.658  0.389
3_4_WR_pass_rate       -0.254 -0.247  0.090 -0.044  0.029
```

Coding Chunk 2: PCA loadings

```
set.seed(123)
k3 <- kmeans(cluster_input, centers = 3, nstart = 50)

round(k3$centers, 3)

```

```
      PC1    PC2    PC3
1 -2.111  1.731  0.389
2  1.225  0.290 -0.095
3 -0.939 -1.609 -0.067
```

Coding Chunk 3: K means centers

```r
fig <- plot_ly(
  plot_data,
  x = ~PC1,
  y = ~PC2,
  z = ~PC3,
  color = ~Cluster,
  colors = c("#1f77b4", "#FF0000", "#2ca02c"),  # Blue, Red, Green
  text = ~paste(
    "Team:", Team,
    "<br>OC:", OC,
    "<br>PC1:", round(PC1, 2),
    "<br>PC2:", round(PC2, 2),
    "<br>PC3:", round(PC3, 2)
  ),
  hoverinfo = "text",
  type = "scatter3d",
  mode = "markers",
  marker = list(size = 6)
)

fig <- fig %>% layout(
  title = "OC Clusters – Interactive PCA (PC1, PC2, PC3)",
  scene = list(
    xaxis = list(title = "PC1 – Pass Identity"),
    yaxis = list(title = "PC2 – Spread vs Heavy"),
    zaxis = list(title = "PC3 – Tempo / Agg")
  )
)

fig
```

Coding Chunk 4: Interactive Plot